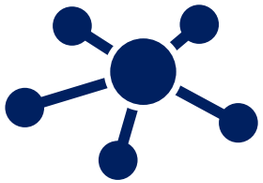
Computer Networking



谢 逸

中山大学·计算机学院

2024. Fall



学术经历

教学科研: 中山大学计算机学院

访问学者: **Deakin U, Melbourne, Australia.**

Georg Mason U, Fairfax, USA.

本/硕/博： 中山大学

联系方式

计算机学院A425

https://cse.sysu.edu.cn/content/2462

+8613710287197

xieyi5@mail.sysu.edu.cn

xieyi-cn or 13710287197 or

- 教学:
  - 本科：计算机网络、通信原理；
  - 硕士/博士：高级计算机网络、高级网络与信息安全

- 科研:
  - 方向：面向计算机网络的基础性研究。
  - 目标：探索通信网络的工作机理及潜在规律，为下一代网络设计、管理及提升通信服务质量提供理论支持。
  - 项目：国家自然科学基金、华为合作项目
  - 成果：网络领域顶刊



研究工作



计算机网络体系结构、
通信协议、网络安全、
资源调控、质量保障、
用户体验、……

算力调度、工业控制、
无人系统、移动计算、
远程教育、智慧城市、
……

欢迎参与网络研究！
欢迎垂询更多细节！

xieyi5@mail.sysu.edu.cn

xieyi-cn or 13710287197 or

《计算机网络》课程概要

- **Prerequisite Courses**
  - **Operating System**
  - **Probability and Statistics**
  - **Programming**

- **Successive Courses**
  - **Principle of communication**
  - **Network Security**
  - **Wireless communication**
  - **…**

《计算机网络》课程概要

章节	主要教学内容	学时	重点、难点
第1章 计算机网络和因特网	课程简介，网络边缘，网络核心，分组交换中的时延、丢包和吞吐量，面对攻击的网络，计算机网络和因特网的历史	6	重点：协议的概念，参考模型，协议层次，各层设计问题 难点：网络相关概念的理解，如协议、服务、交换等，计算机网络的体系结构及各层交互的封装过程
第2章 应用层	应用层协议的原理，应用层协议的实现过程，进程通信，HTTP协议的持续连接，Web应用和HTTP协议，FTP协议，电子邮件协议，DNS协议，P2P文件共享，TCP和UDP套接字编程基础，C/S架构应用实例	8	重点：应用层协议的原理，Web应用和HTTP协议，TCP和UDP套接字编程 难点：HTTP协议的持续连接，应用层协议的实现过程，DNS协议，P2P协议
第3章 传输层	传输层的基本概念和服务，多路复用和多路分解，UDP的实现方法和相关传输原理，可靠数据传输原理，TCP连接，TCP拥塞控制	10	重点：多路复用和多路分解、TCP协议 难点：TCP的拥塞控制
第4章 网络层：数据平面	网络层工作原理，网络层的转发，IP协议，网络服务和模型，通用转发和SDN	6	重点：网络层工作原理 难点：IP协议与数据包的网络层转发
第5章 网络层：控制平面	路由选择算法，OSPF协议，BGP协议，SDN的控制平面，ICMP协议，网络管理和SNMP协议	8	重点：路由选择算法 难点：路由协议
第6章 链路层和局域网	链路层服务，链路层协议，差错检验方法，CRC校验，滑动窗口协议，多路访问协议，链路层原理，ARP协议，以太网技术	10	重点：多路访问协议，CRC校验，以太网的工作原理 难点：多路访问协议，以太网的工作原理
第7章 无线网络	无线链路与网络特征，WiFi:802.11，无线局域网	4	重点：WiFi:802.11协议，无线局域网 难点：WiFi:802.11协议
考试	通过考试、进一步强化学生对计算机网络基础知识的理解和掌握	2	无

《计算机网络》课程概要

课程环节安排：

- 授课：采用多媒体辅助课堂教学。
- 实验：课堂教学与另外单独设置的计算机网络实验课程相配合。
- 作业：每章均有作业。
- 考试：期末闭卷考试，内容涵盖课堂上所讲授的内容，包括基本概念、计算和分析等题目。

《计算机网络》课程概要

教学方法：

- 课堂授课：以课堂讲解基本概念和原理为主，作业为辅。
- 多媒体教学：利用**PPT**呈现教学内容，提高教学效率。
- 课堂讨论：对教学内容的重点，通过课堂讨论加深理解。
- 作业讲解：讲解作业问题，总结作业情况。
- 答疑：每周**2**小时办公时间答疑。
  - 作业：直接联系助教；
  - 理论：每周三中午（**12**点**-14**点，学院楼**A425**），请提前约定；

《计算机网络》课程概要

教材与参考书

- 教材名称：计算机网络：自顶向下方法(原书第8版)
  - 作者：[美] **James F.Kurose，Keith W.Ross** 著，陈鸣 译
  - 出版社：机械工业出版社

- 主要参考书目
  - 《计算机网络》第8版，谢希仁编著， 电子工业出版社, **2021.**
  - 《**Computer Networks**》**5th edition**(英文版), **A. S. Tanenbaum**，清华大学出版社，**2011.**

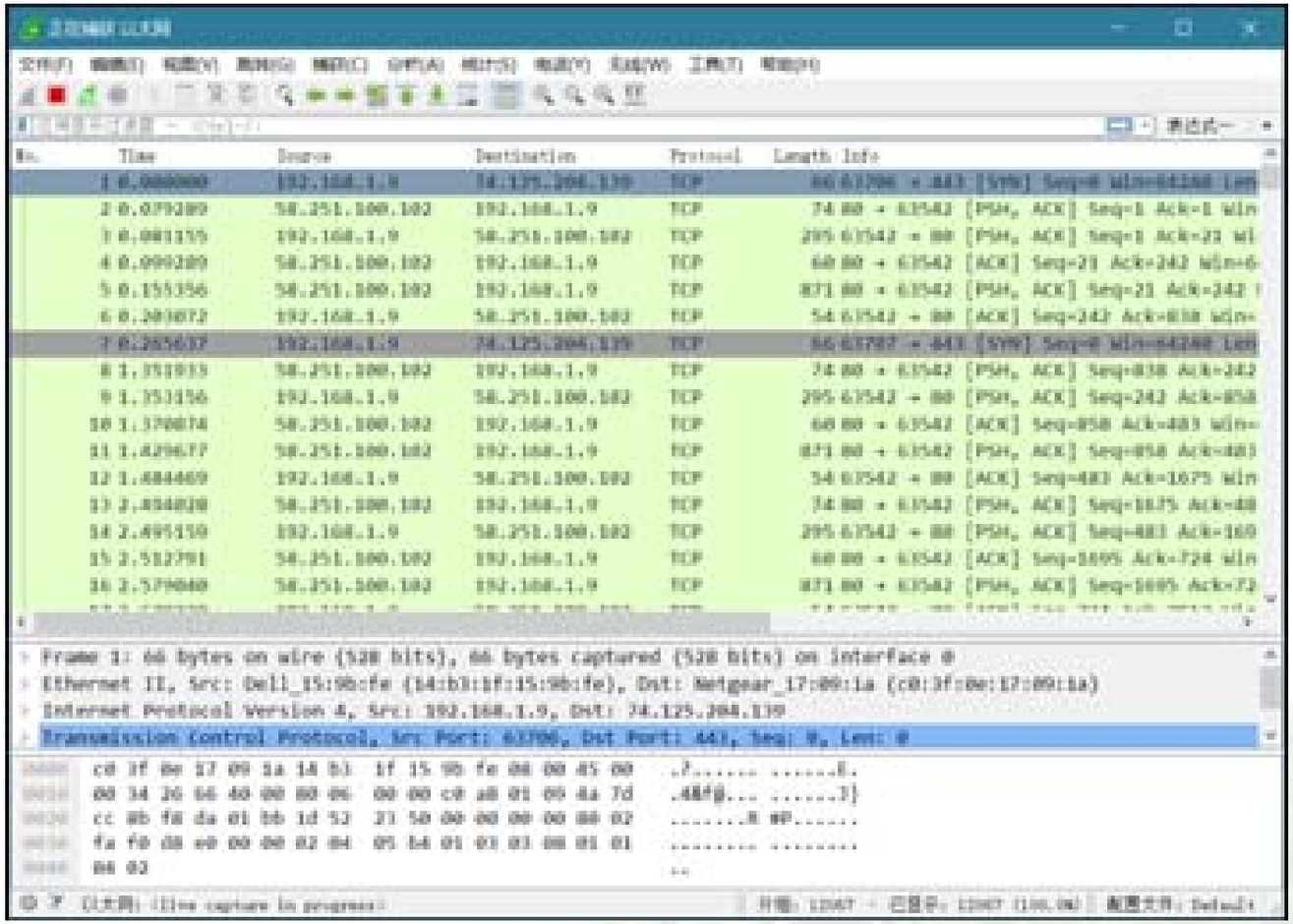
《计算机网络》课程概要

成绩评定方式

- 考 核 方 式：考试
- 成绩录入方式： 百分制
- 期末考试：闭卷，占总成绩比例**60%**。
- 平时成绩：书面作业，约**5-7**次，占总成绩比例**35%**；随堂考勤约**5**次，占总成绩比例**5%**。

常用工具

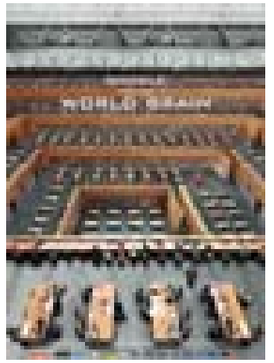
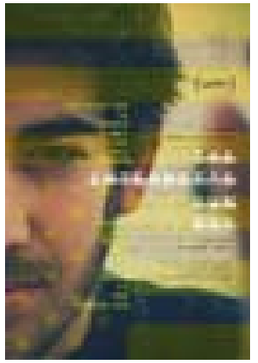
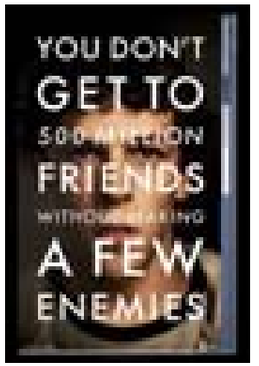
- **Experimental tool: Wireshark**
  - **Wireshark is a free and open source packet analyzer. It is used for network troubleshooting, analysis, software and communications protocol development, and education.**
  - **https://www.wireshark.org/**



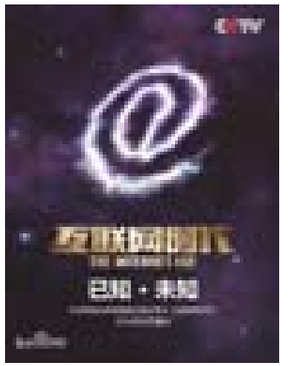
课外读物

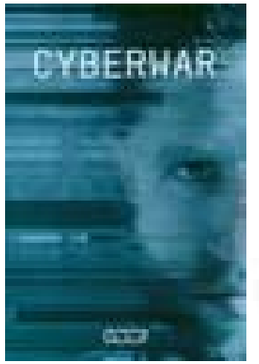
- **Recommended video:**
  - **BBC: “Virtual Revolution”**
  - **BBC: “Google and the World Brain”**
  - **“The Internet’s own boy”**
  - **“The Social Network”**
  - **“维基解密的抗争”**
  - **CCTV: “互联网时代”**

- **Cyberwar season 1**
- **Cyberwar threat**
- **…**











Miscellaneous

- Schedule: 18 Weeks vs. 500 PPTs ~~ 50 pages/week
- Teaching pattern:
  - in class: We will only focus on key points, and encourage students to question and discuss some interesting topics.
  - after class: Students have to read, think and prepare the issues for class discussion. Otherwise, it will be difficult for passing the tests.
  - My advice:
    - Please show your questions in the class;
    - Please focus on “Why”, “How”;
    - please do NOT rely on your memory!

What can you get from this course?



- Basics of Data Communications;
- Ability to observe, think and solve problems. (More important)

Some things about the experimental class

- Experimental classes start from the first week；
- Please complete the grouping for the experimental class before this Wed.: 4 students per group
  - An online table will be created for it and distributed via Enterprise WeChat.

Contact Information

- **Teaching Assistant:** 汪昊元
- 课程微信群(企业微信):

●理论课PPT讲义：

https://gaia.cs.umass.edu/kurose_ross/index.php

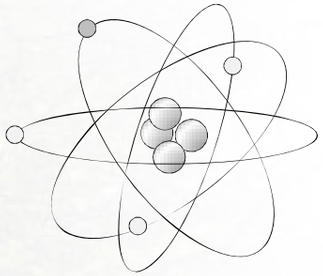
https://pan.sysu.edu.cn/link/AABA104C02E83E473A9BFB2E7E3DE8F96A
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有效期限：2025-01-20 21:30
提取码：sLsT

第1节 内容：

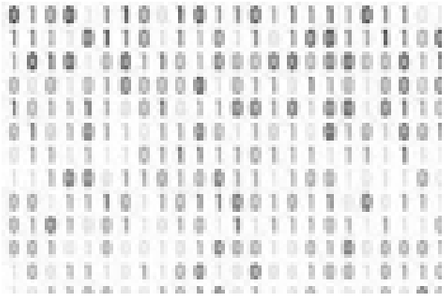
- 开设计算机网络课程的原因
- 现代数据网络的发展历史

为什么需要学习计算机网络?

- 人类历史上影响最深远的发明创造之一；
- 互联网促使现实物理世界和虚拟数字世界走向深度融合。

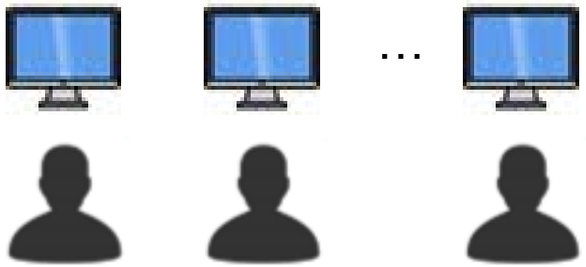






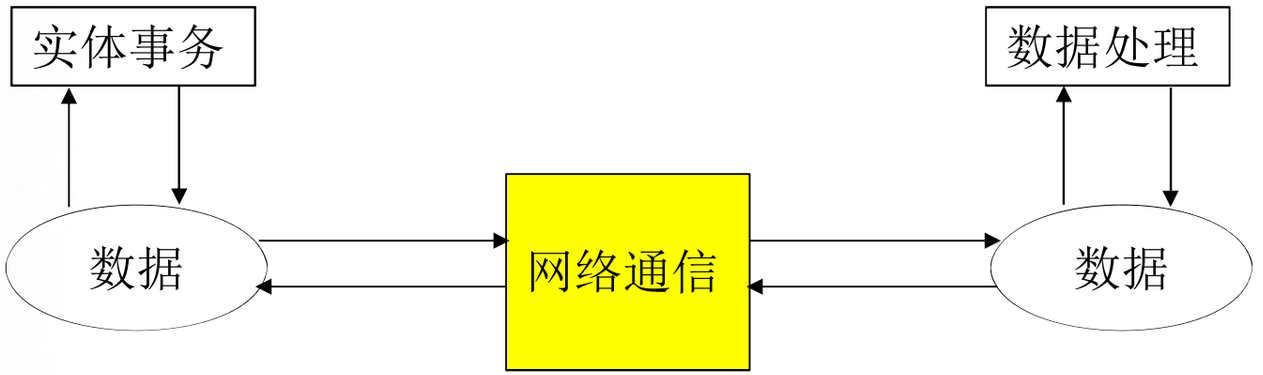
实体世界　　虚拟世界







- 物理世界中的事务处理方式转向以数据为基础、通信为支撑的模式。
- 网络空间全球一体化成为不可逆转的发展方向。



网络是全球不可或缺的关键基础设施之一

- 网络是连接世界的重要手段
- 网络是信息传递的唯一途径
- 通信深植于现代日常生活中
- 涉及个人隐私、社会稳定、国家安全





移动蜂窝、卫星通信、无人机通信、水声通信、可见光通信 …

高速率、低时延、高可靠、高效率、高能效、…

人、物、智能体、…

空、天、地、海、…

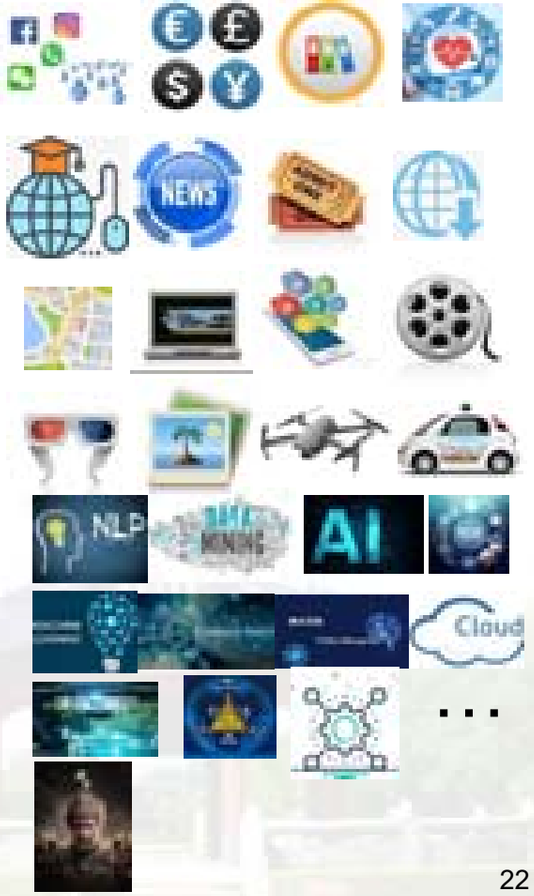
数据 网络
Si 60s

生活、经济、科技、文化、国防、管理、生产、…

跨地域、跨空域、跨海域、空-天-海-地-体化、…

连接物理与虚拟边界，人-机-物-境的协作

一体化的网络智能及内生安全策略



- 网络不可用时会怎么样？
  - 当网络不可用时：

数据无法传递	消息无法发送	网络资源失联	电商不可访问	游戏无法继续
视频无法观看	图像识别失效	NLP停止工作	机器人罢工	人工智能停顿
无人机停飞	直播无法上网	金融交易中断	远程手术中断	远程会议中断
工控国防瘫痪	影响人类生活	影响社会稳定	威胁国家安全	……

在所有IT相关的技术应用中，没有一种具有计算机网络一样的战略地位！

数据通信网络面临的挑战

- 网络业务与应用快速更新迭代；
  - 多媒体、**VR**、游戏；
  - 人工智能、分布式算力；
  - 教育、医疗、金融、国防；
  - 工业、无人系统、物联网；
  - ……
- 网络体系结构不适应现代需求；
  - 通信质量需求；
  - 用户体验需求；
  - 数据业务需求；
  - 通信安全需求；
  - ……

网络技术需要与时俱进

6G（2030）

现代网络的发展简史

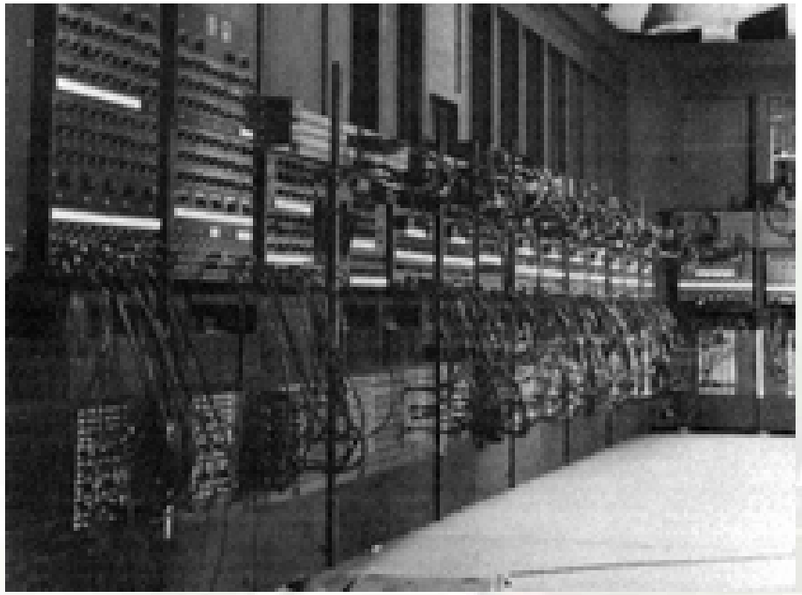
- **Part I Connecting Hosts**
- Part II Connecting Networks
- Part III Connecting the World
- Part IV Connecting everything

了解过去、认识现在、创造未来

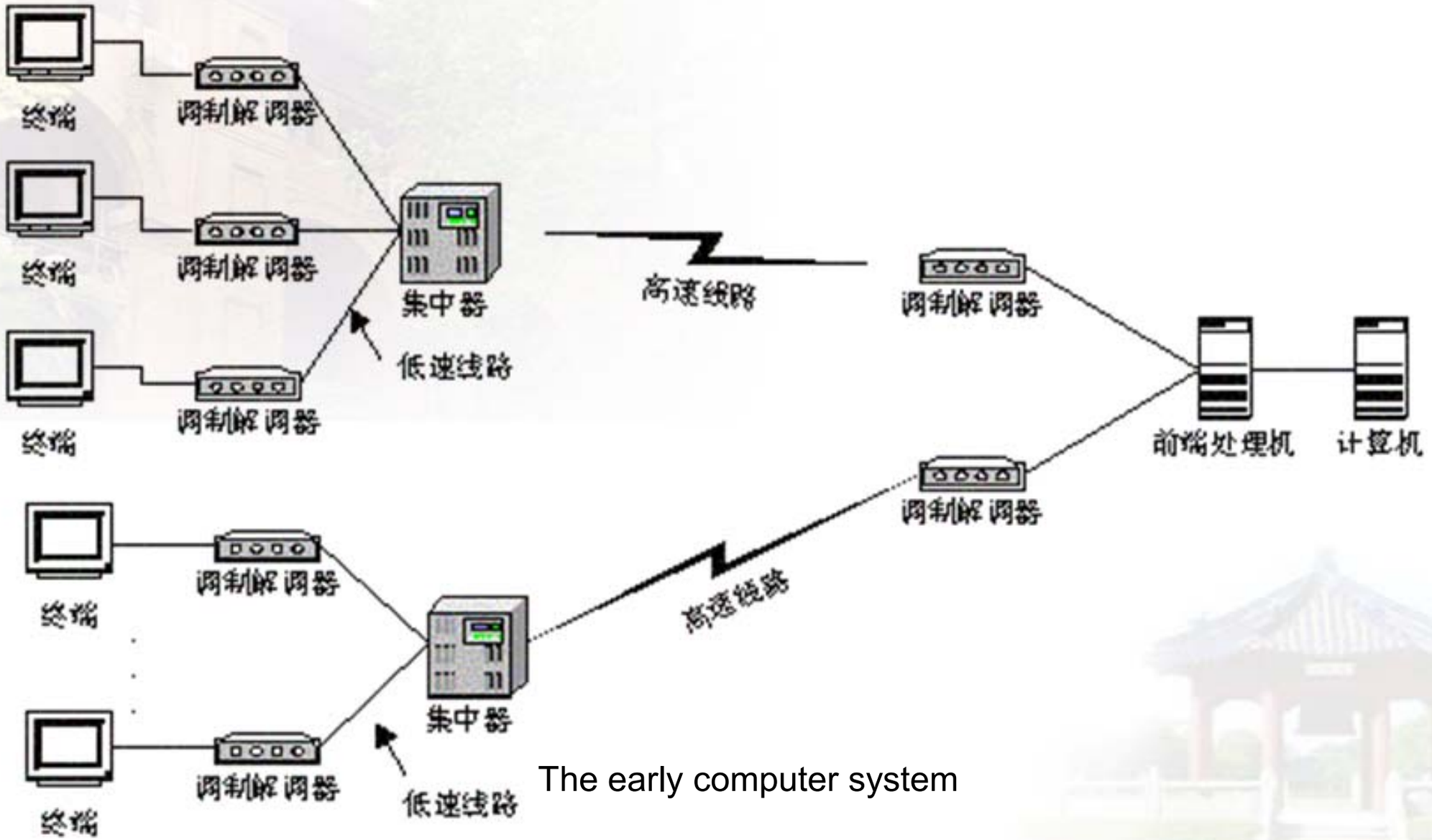
Connecting Hosts (1940-1972)

- 1940, George Stibitz used a Teletype machine to send instructions or a problem set from his Model at Dartmouth College to his Complex Number Calculator in New York and received results back by the same means.
- 1946, The first electronic computer, ENIAC, was born in the United States.
- For the next 20 years people have been studying the fusion of computers and communication systems.



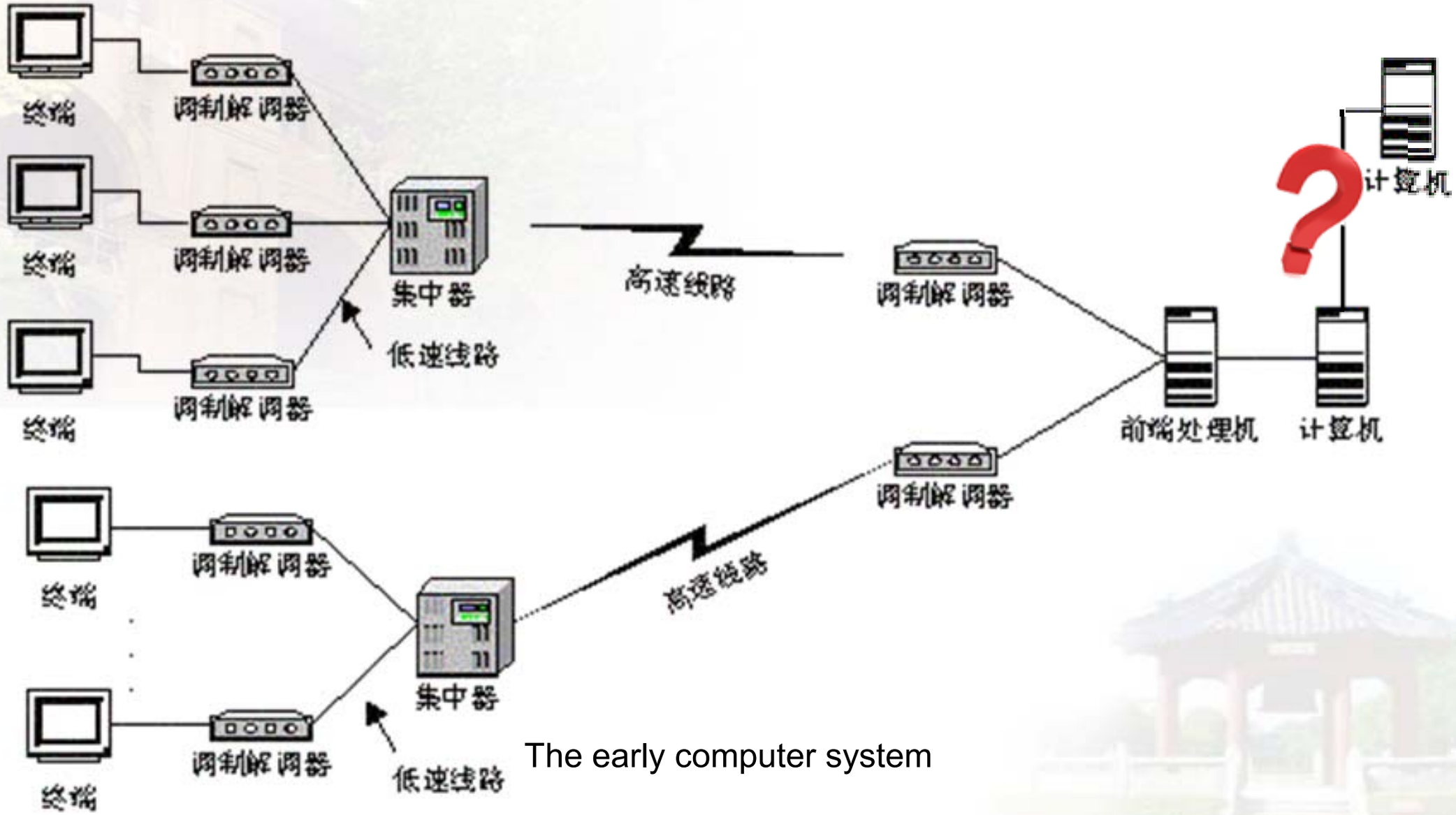


Connecting Hosts (1940-1972)



The early computer system

Connecting Hosts (1940-1972)



The early computer system

• Connecting Hosts (1940-1972)



The story began in the 1950s. It is a star-studded age in which there is a group of Internet pioneers admired by future generations.

INTERNET HALL of FAME PIONEERS



Paul Baran
(Packet switching)



Vint Cerf
(TCP IP, IP Standards, DARPA, Father of the Internet, ARPANET)



Steve Crocker
(TCP IP, IP Standards, ICANN, ARPANET)



Robert Kahn
(TCP IP, IP Standards, DARPA, Father of the Internet, ARPANET)



Donald Davies
(Packet switching, ARPANET)



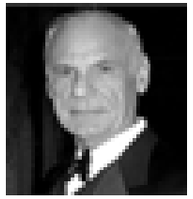
Danny Cohen
(Packet switching, ARPANET)



Leonard Kleinrock
(Packet switching, IP Standards, ARPANET)



J.C.R. Licklider
(Intergalactic Computer Network, Internet, ARPA, ARPANET)



Lawrence Roberts
(ARPANET)



Craig Newmark
(web-based platform)



Linus Torvalds
(Open Source Internet, UNIX)



Jon Postel
(Request for Comment, TCP IP, IP Standards, DNS, Internet Assigned Authority, ARPANET)



Tim Berners-Lee
(World Wide Web, CERN)



Raymond Tomlinson
(TELNET, ARPANET)



Bob Metcalfe
(Ethernet, Xerox PARC, MIT, 3Com, TCP/IP, Hewlett Packard, ARPANET)



Mitchell Baker
(Open Source Internet, World Wide Web)

. . .

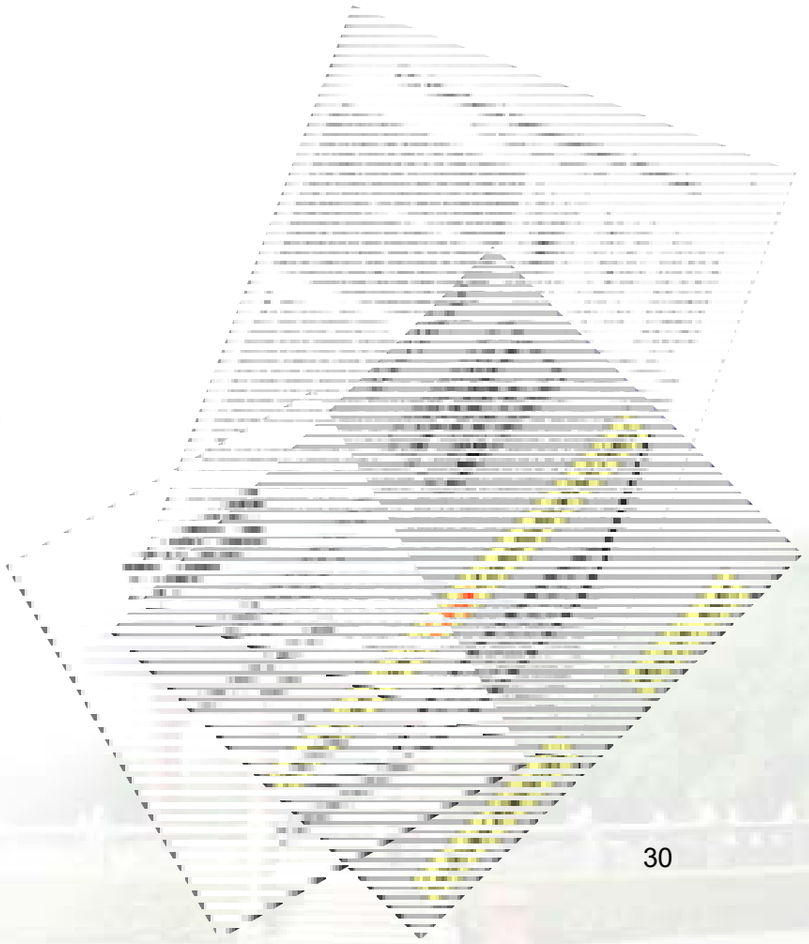
https://www.internethalloffame.org



Connecting Hosts (1940-1972)

1957,	Sputnik launched
1958, Beginning	Advanced Research Projects Agency (ARPA) in the Department of Defense (DoD) was formed as response

- Research on packet networks during 1959-1966:
  - Lenoard Kleinrock: *Information Flow in Large Communication Networks*, PhD Thesis, 1961.
  - Paul Baran: "*On Distributed Communications Networks*", 1961.
  - Larry Roberts/Tom Marill publish first paper on network experiments: "*Toward a Cooperative Network of Time-shared Computer*" 1966.



- **Connecting Hosts (1940-1972)**
  - **The motivation of connecting terminals:**

I. … consider the synthesis of a communication network which will allow several hundred major communications stations to talk with one another after an enemy attack…

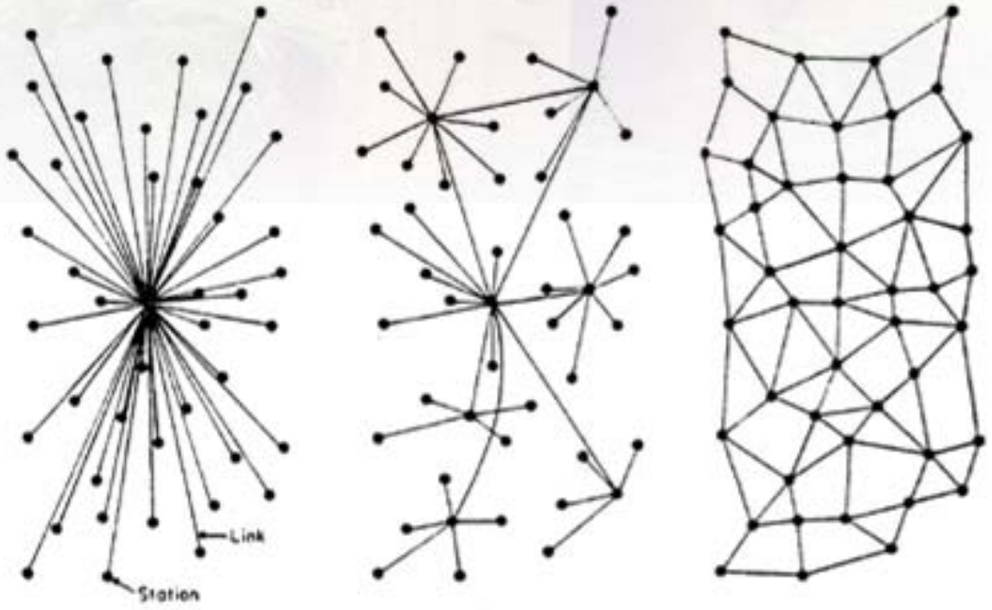


Fig. 1—(a) Centralized. (b) Decentralized. (c) Distributed networks.

… we must use a switching scheme able to find any possible path that might exist after heavy damage … we shall explore the possibilities of building a "real-time" data transmission system using store-and-forward techniques…

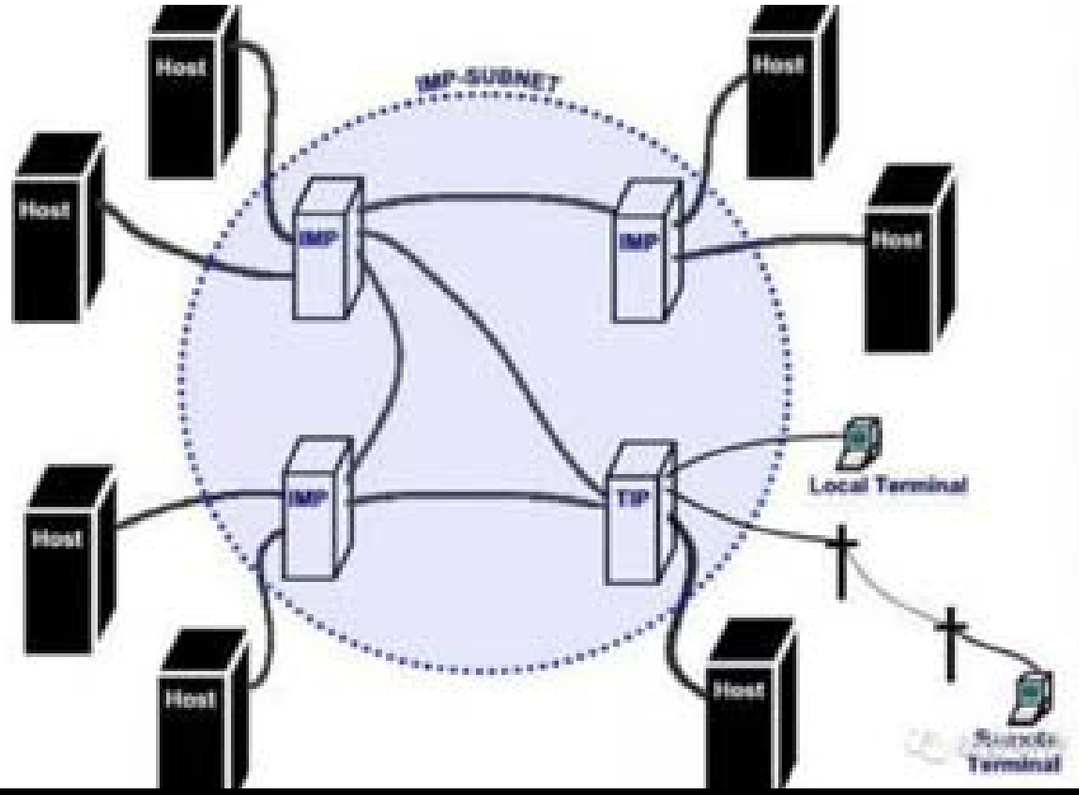
Paul Baran, "On distributed communications networks," 1962

This directly derived the packet switching and store-and-forward.

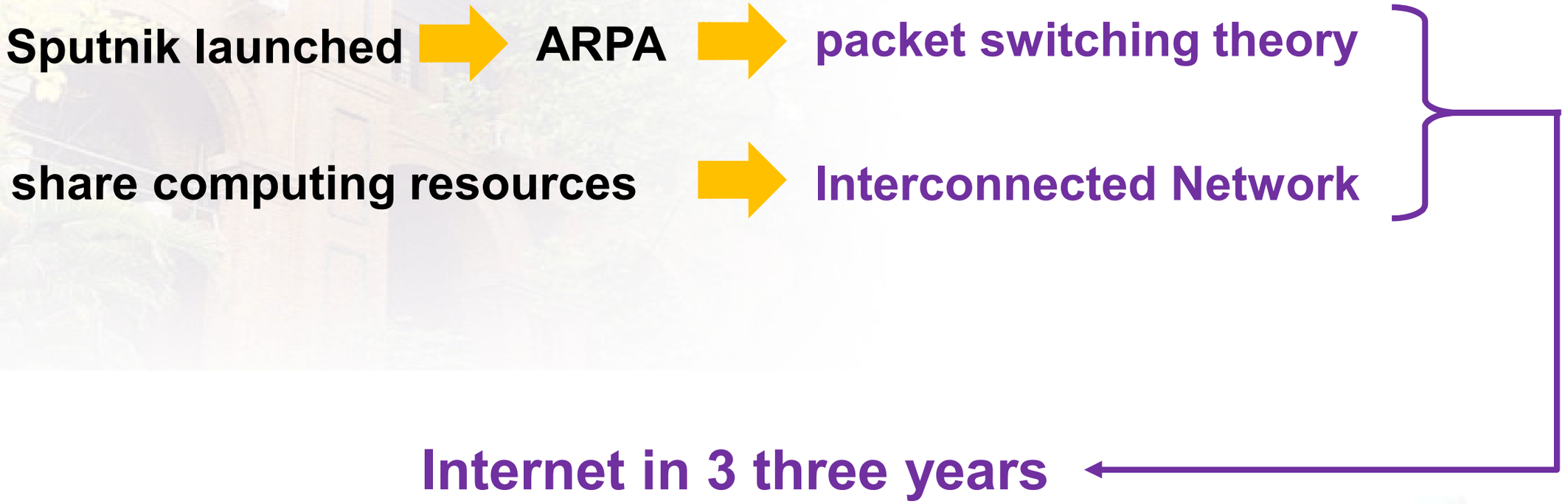
- **Connecting Hosts (1940-1972)**
  - **The motivation of connecting terminals:**

II. In 1966, Robert Taylor decided that ARPA should link the existing computers at ARPA-funded research institutions together. This would allow everybody on the network to **share** computing resources and results. Larry Roberts ("father of the ARPANET"), managed the project.

This directly derived the ARPA net.



- **Connecting Hosts (1940-1972)**



• Connecting Hosts (1940-1972)

■ **1959-62, a mathematical theory of packet networks is created at MIT by Leonard Kleinrock**



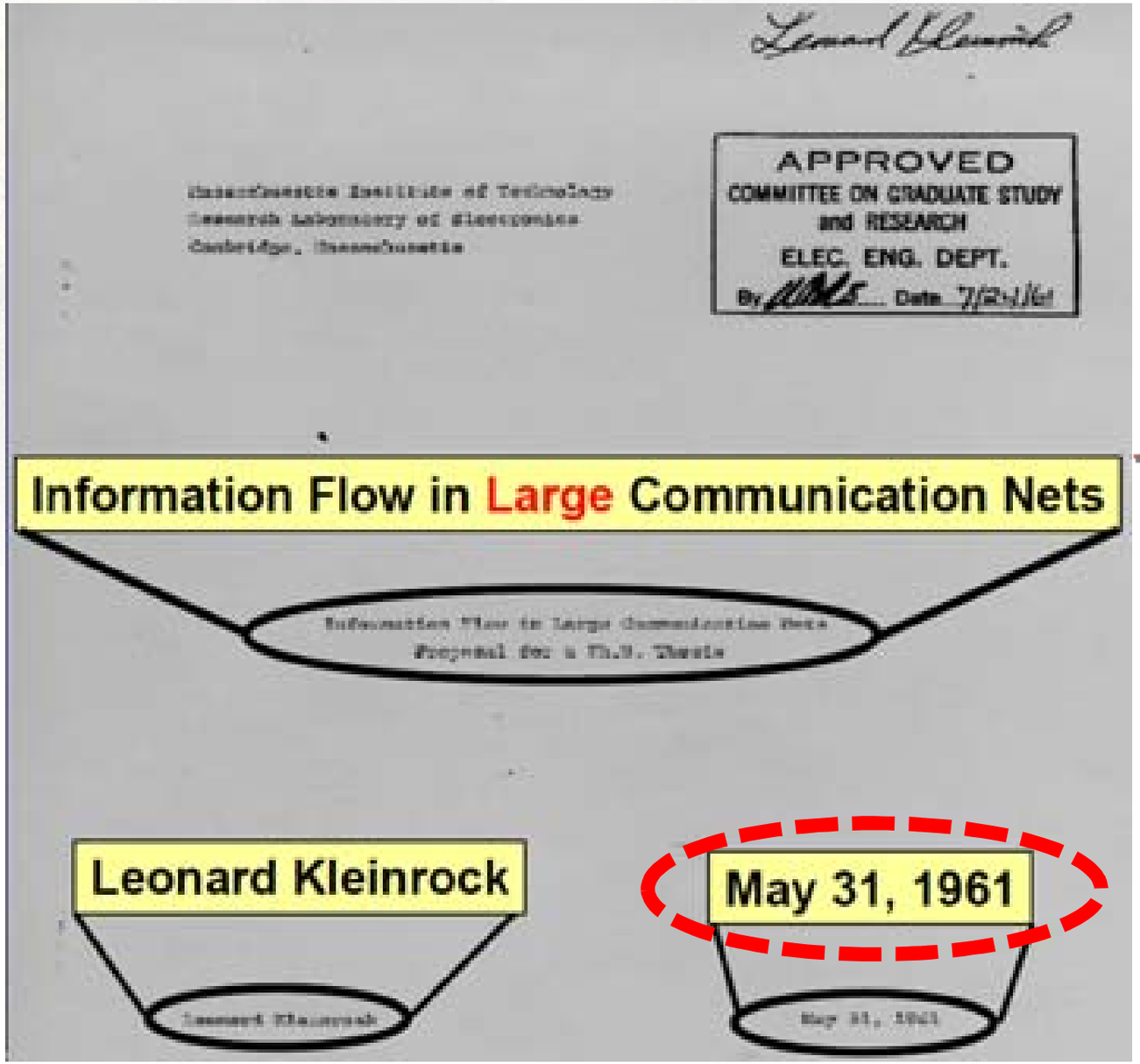
Professor Leonard Kleinrock is Distinguished Professor of Computer Science at UCLA. Known as a "**Father of the Internet**," he developed ***the mathematical theory of packet networks***, the technology underpinning the Internet, while a graduate student at MIT. This was in the period 1960–1962, nearly **a decade before the birth of the Internet** which occurred in his laboratory when his Host computer at UCLA became the first node of the Internet in September 1969. He wrote **the first paper** and published **the first book** on the subject; he also directed the transmission of **the first message ever to pass over the Internet**.

Internet Hall of Fame

Internet Society WIRED

https://www.computer.org/profiles/leonard-kleinrock

- **Connecting Hosts (1940-1972)**



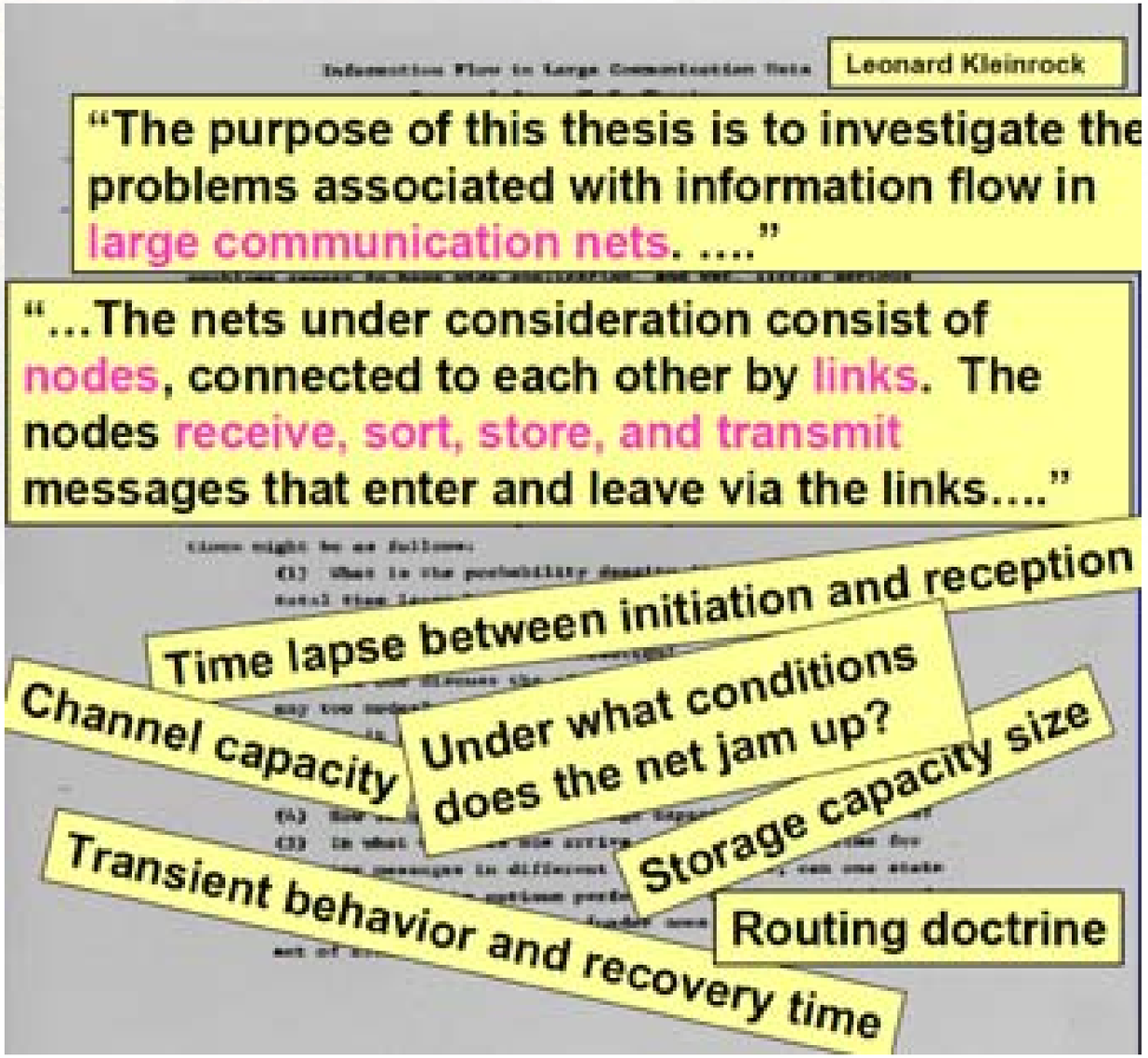
Leonard Kleinroch's Ph.D thesis:

Information flow in large communication nets

Packet switching theory is proposed

The idea, concept, implementation and technical details of packet switching were first proposed.

• Connecting Hosts (1940-1972)



Keywords (1961):

- Information flow,
- Large communication nets,
- nodes,
- links,
- receive,
- transmit,
- store,
- sort,
- storage capacity,
- Channel capacity,
- jam up,
- recovery,
- routing,
- transient behavior,
- …

• Connecting Hosts (1940-1972)

The precedure for servicing is as follows: A unit upon srrival joins the end of the queue, and waits on line in a first come first served fashion until finally arrives at the service facility. The server picks the next unit in the queue and performs a unit of service upon it. At the end of this time interval, the unit leaves the system if its service is finished; if not, it joins the end of the queue with its service partially completed.

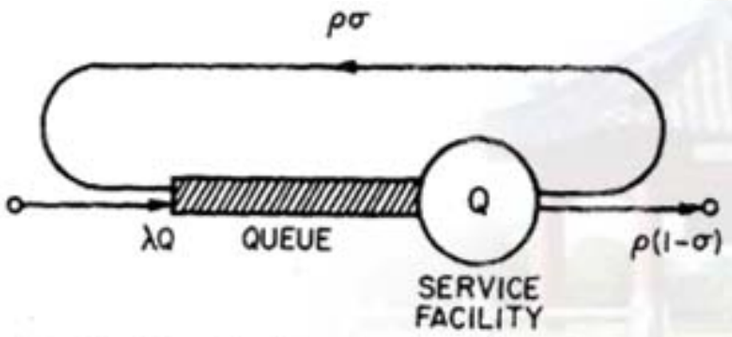


FIG. 1. Round-robin time-shared service system

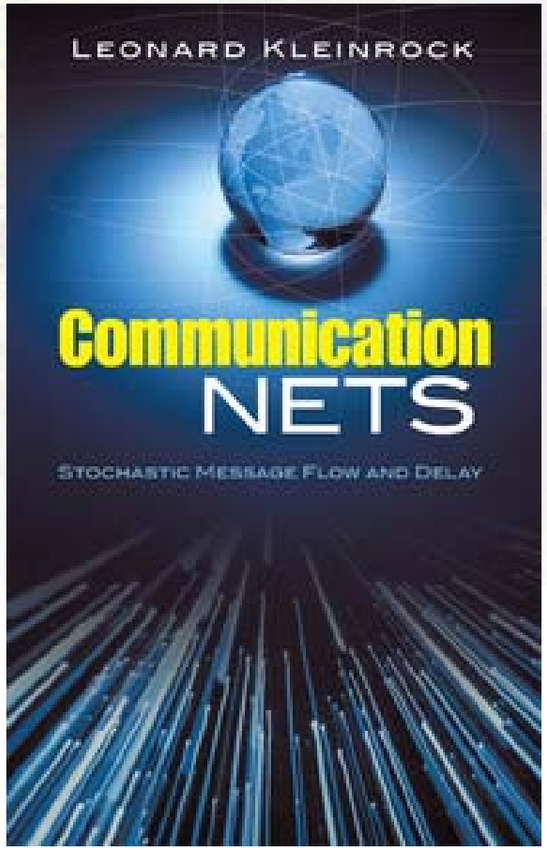
• Connecting Hosts (1940-1972)

Leonard Kleinrock

Created a mathematical theory of data networks

- **Channel capacity limited**
- **Mean response time as key metric**
- **Optimal assignment of channel capacity**
- **Choice of priority queueing discipline**
- **Concept of breaking messages into fixed size blocks**
- **Choice of routing procedure**
- **Design of topological structure**
- **Developed underlying principles of data networks that are the basis of the Internet**

• Connecting Hosts (1940-1972)



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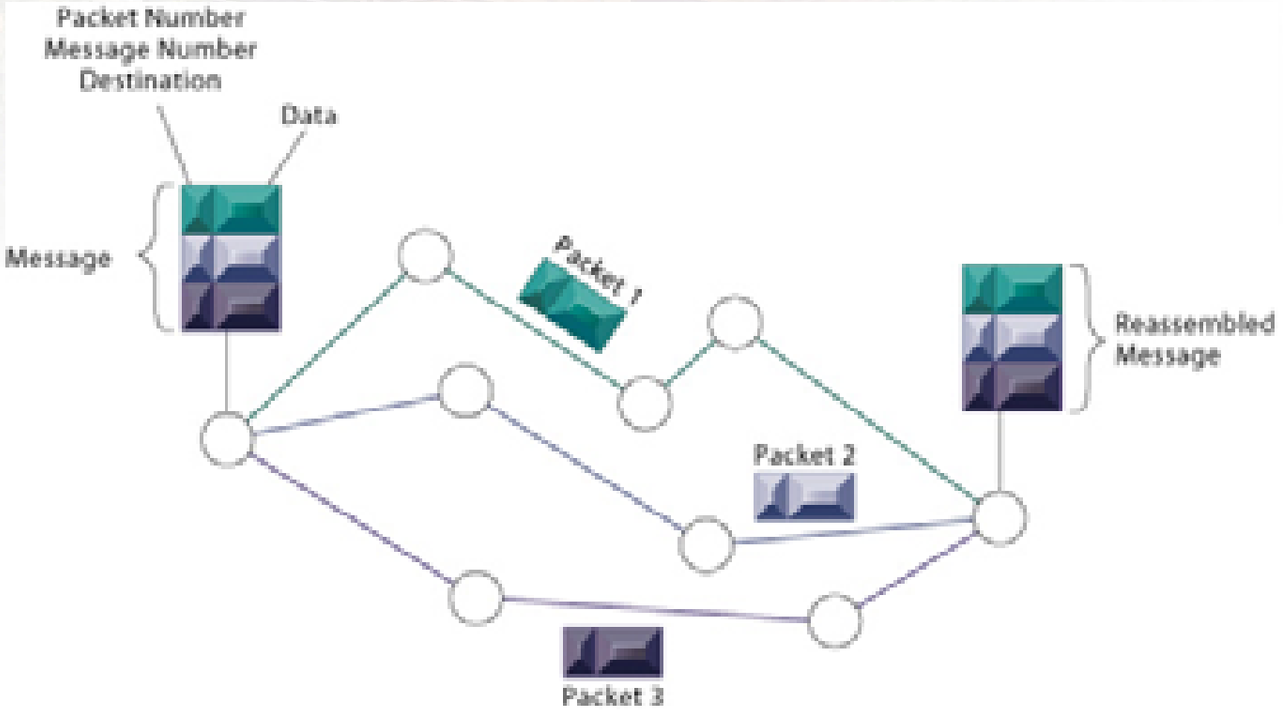
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Communication nets: stochastic message flow and delay
By: Leonard Kleinrock
1964

• Connecting Hosts (1940-1972)

- 1962, Pau Baran suggests transmission of data using fixed size message blocks
- 1964, Pau Baran publishes reports “on distributed communications”

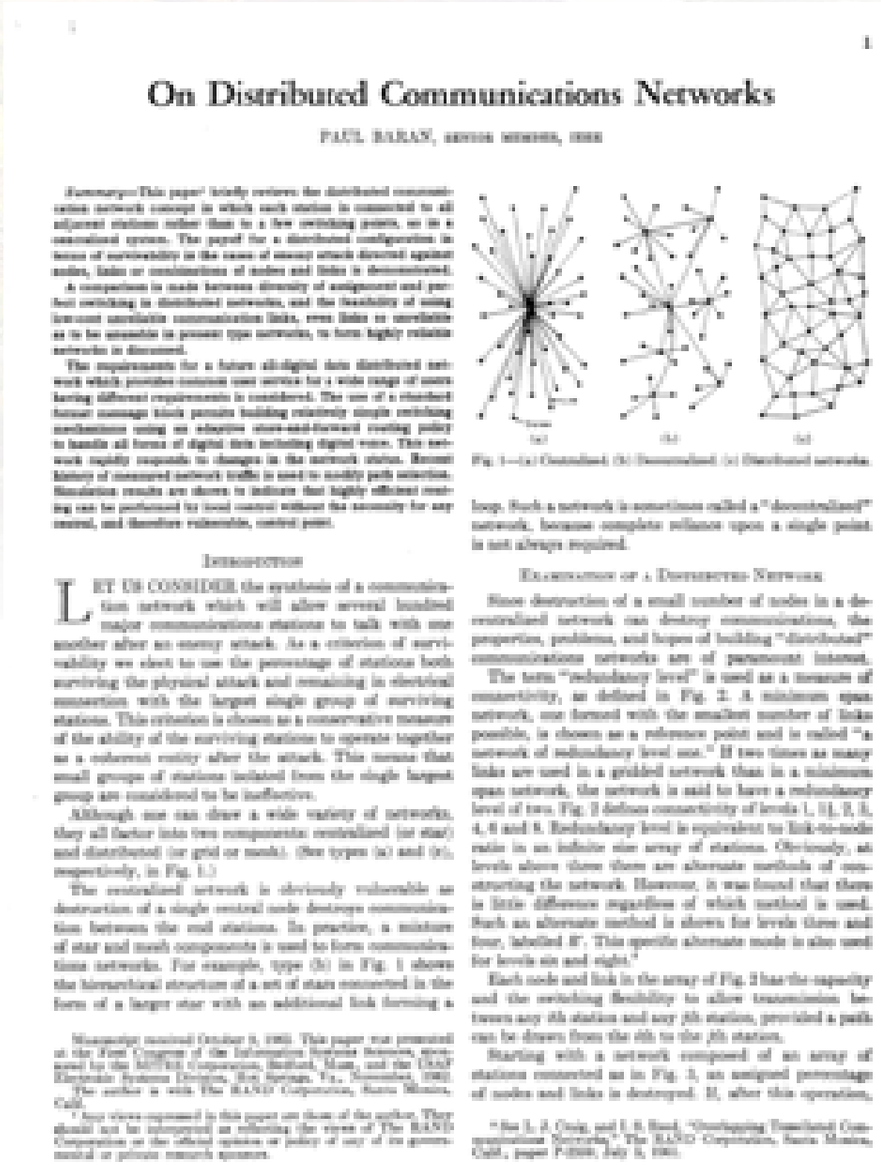




packetization

he worked on developing a communications system that could survive the damage of a nuclear weapon. Originally, he called the process “message blocks.” Other scientists including Donald Davies, who would later change the name to “packet switching,” had also come to a similar conclusion at the same time.

Connecting Hosts (1940-1972)



On Distributed Communications Networks

PAUL BARAN, SENIOR MEMBER, IEEE

Summary—This paper briefly reviews the distributed communication network concept in which each station is connected to all adjacent stations rather than to a few switching points, as in a centralized system. The payoff for a distributed configuration in terms of survivability in the cases of enemy attack directed against nodes, links or combinations of nodes and links is demonstrated.

A comparison is made between diversity of assignment and perfect switching in distributed networks, and the feasibility of using low-cost unreliable communication links, even links so unreliable as to be unusable in present type networks, to form highly reliable networks is discussed.

The requirements for a future all-digital data distributed network which provides common user service for a wide range of users having different requirements is considered. The use of a standard format message block permits building relatively simple switching nodes which can serve all users. The use of adaptive store-and-forward routing policy is tested to handle all forms of digital data including digital voice. This network rapidly responds to changes in the network status. Recent history of measured network traffic is used to modify path selection. Simulation results are shown to indicate that highly efficient routing can be performed by local control without the necessity for any central, and therefore vulnerable, control point.

Fig. 1—(a) Centralized. (b) Decentralized. (c) Distributed networks.

Introduction

LET US CONSIDER the synthesis of a communication network which will allow several hundred major communications stations to talk with one another after an enemy attack. As a criterion of survivability we elect to use the percentage of stations both surviving the physical attack and remaining in electrical connection with the largest single group of surviving stations. This criterion is chosen as a conservative measure of the ability of the surviving stations to operate together as a coherent entity after the attack. This means that small groups of stations isolated from the single largest group are considered to be ineffective.

Although one can draw a wide variety of networks, they all factor into two components: centralized (or star) and distributed (or grid or mesh). (See types (a) and (c), respectively, in Fig. 1.)

The centralized network is obviously vulnerable as destruction of a single central node destroys communication between the end stations. In practice, a mixture of star and mesh components is used to form communications networks. For example, type (b) in Fig. 1 shows the hierarchical structure of a set of stars connected in the form of a larger star with an additional link forming a loop. Such a network is sometimes called a "decentralized" network, because complete reliance upon a single point is not always required.

Examination of a Distributed Network

Since destruction of a small number of nodes in a decentralized network can destroy communications, the properties, problems, and hopes of building "distributed" communications networks are of paramount interest.

The term "redundancy level" is used as a measure of connectivity, as defined in Fig. 2. A minimum span network, one formed with the smallest number of links possible, is chosen as a reference point and is called "a network of redundancy level one." If two times as many links are used in a gridded network than in a minimum span network, the network is said to have a redundancy level of two. Fig. 2 defines connectivity of levels 1, 1½, 2, 3, 4, 6 and 8. Redundancy level is equivalent to link-to-node ratio in an infinite size array of stations. Obviously, at levels above three there are alternate methods of constructing the network. However, it was found that there is little difference regardless of which method is used. Such an alternate method is shown for levels three and four, labelled B'. This specific alternate mode is also used for levels six and eight.

Each node and link in the array of Fig. 2 has the capacity and the switching flexibility to allow transmission between any ith station and any jth station, provided a path can be drawn from the ith to the jth station.

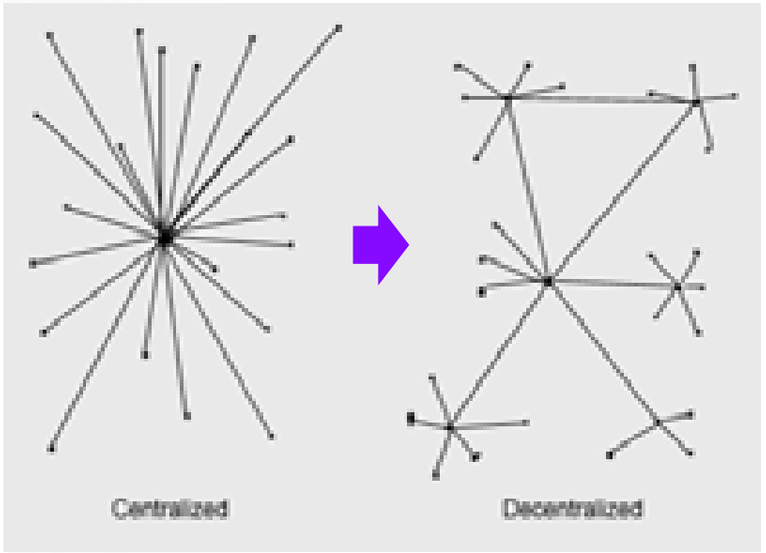
Starting with a network composed of an array of stations connected as in Fig. 3, an assigned percentage of nodes and links is destroyed. If, after this operation,

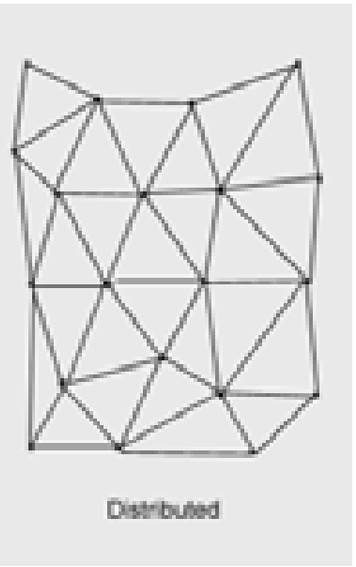
Manuscript received October 6, 1960. This paper was presented at the First Congress of the Information Systems Sciences, sponsored by the MITRE Corporation, Bedford, Mass., and the Air Force Systems Command, Hot Springs, Va., November, 1962.

The author is with The RAND Corporation, Santa Monica, Calif.

* Any views expressed in this paper are those of the author. They should not be interpreted as reflecting the views of The RAND Corporation or the official opinion or policy of any of its governmental or private research sponsors.

† See J. J. Cook and L. E. Reed, "Interhanging Distributed Communications Networks," The RAND Corporation, Santa Monica, Calif., paper P-2306, July 5, 1961.





distributed network

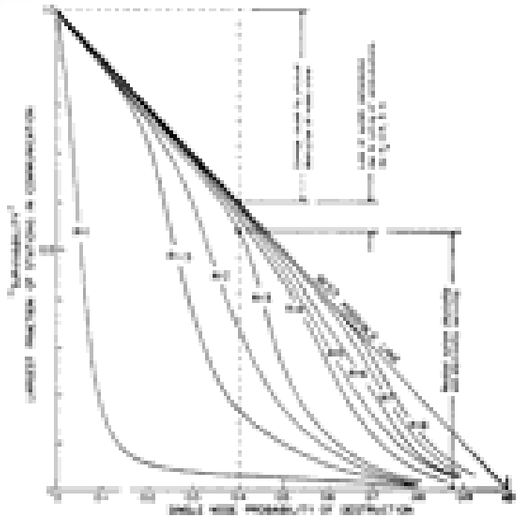


Fig. 4—Perfect switching in a distributed network: sensitivity to node destruction. 100 per cent of links operative.

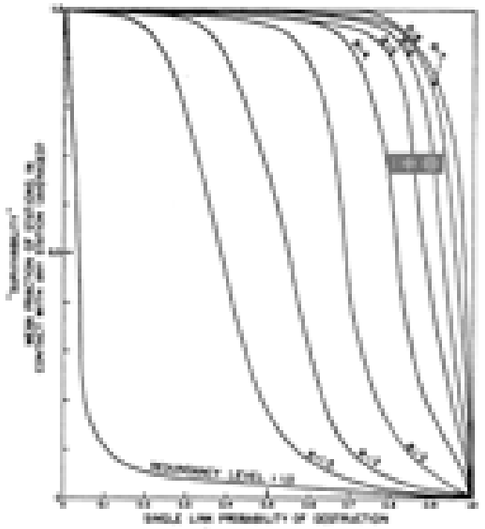
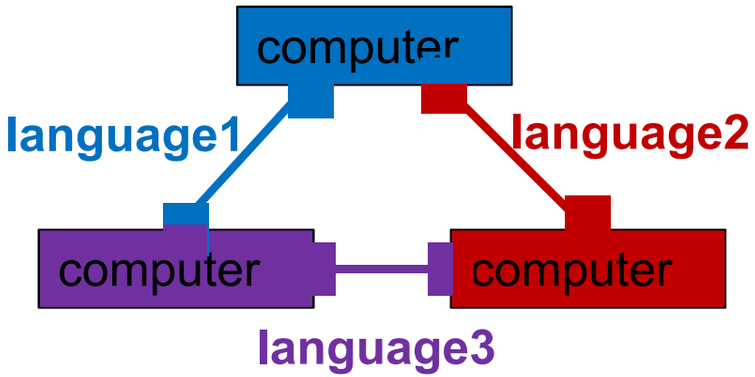
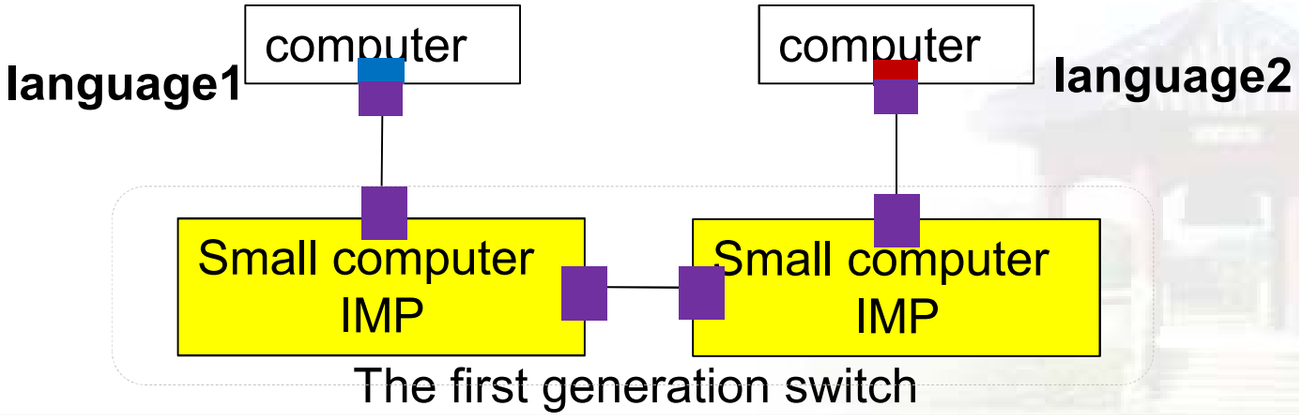


Fig. 5—Perfect switching in a distributed network: sensitivity to link destruction. 100 per cent of nodes operative.

Connecting Hosts (1940-1972)

- Wes Clark suggested that Roberts employ small computers at each site to handle networking functions and leave the host computers alone.
- All of the small computers could speak the same language which would facilitate communication between them.
- Each host computer would only have to adapt its language once in communicating with it's small computer counterpart.
- Each host computer would be connected to the network via its small computer which would act as a sort of gateway.
- The small computers could also remain under more direct ARPA control than were the large host computers.





Connecting Hosts (1940-1972)

- The small computer is called Interface Message Processors (IMPs) in 1967, which became the router/switch later.

The Interface Message Processor (IMP) was the packet switching node used to interconnect participant networks to the ARPANET from the late 1960s to 1989. It was the first generation of gateways, which are known today as routers.

Data from: Wikipedia



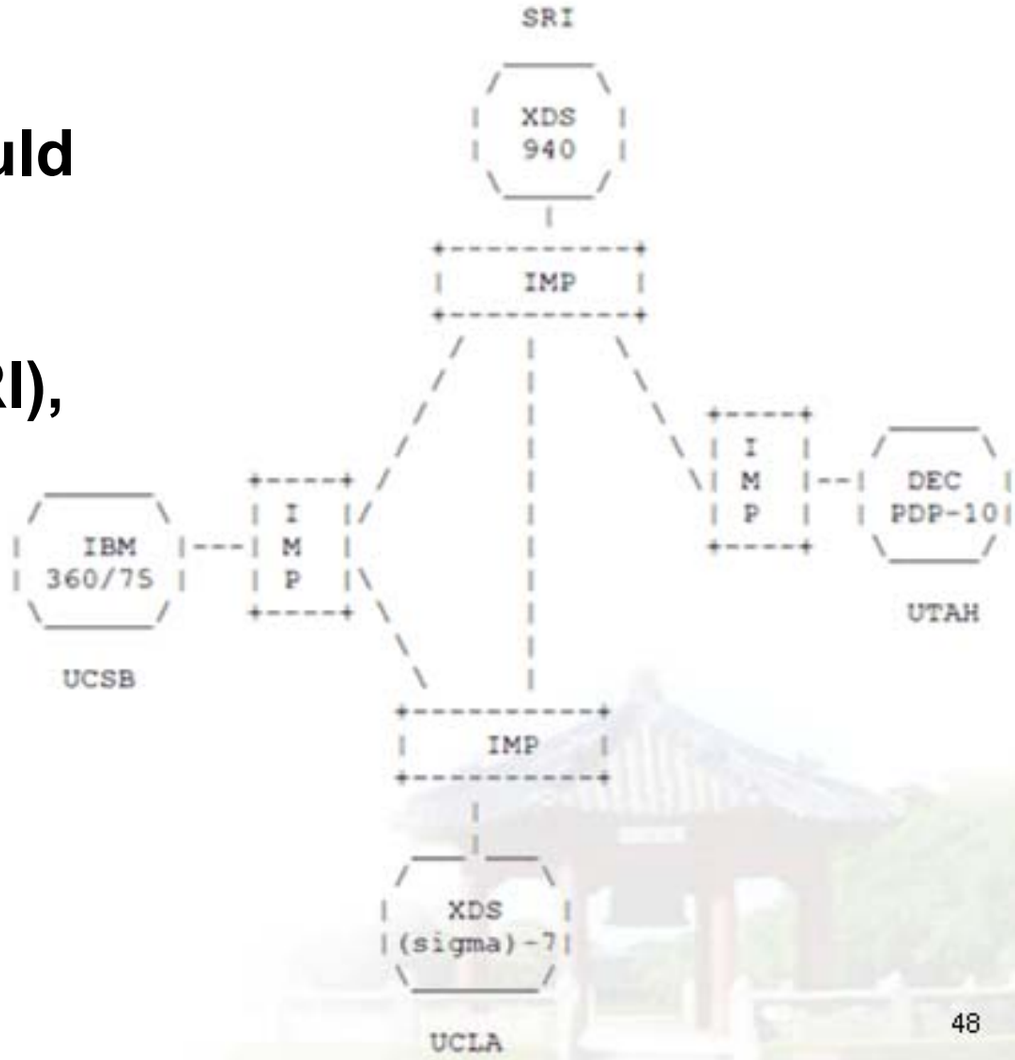
Interface Message Processor (IMP)

Connecting Hosts (1940-1972)

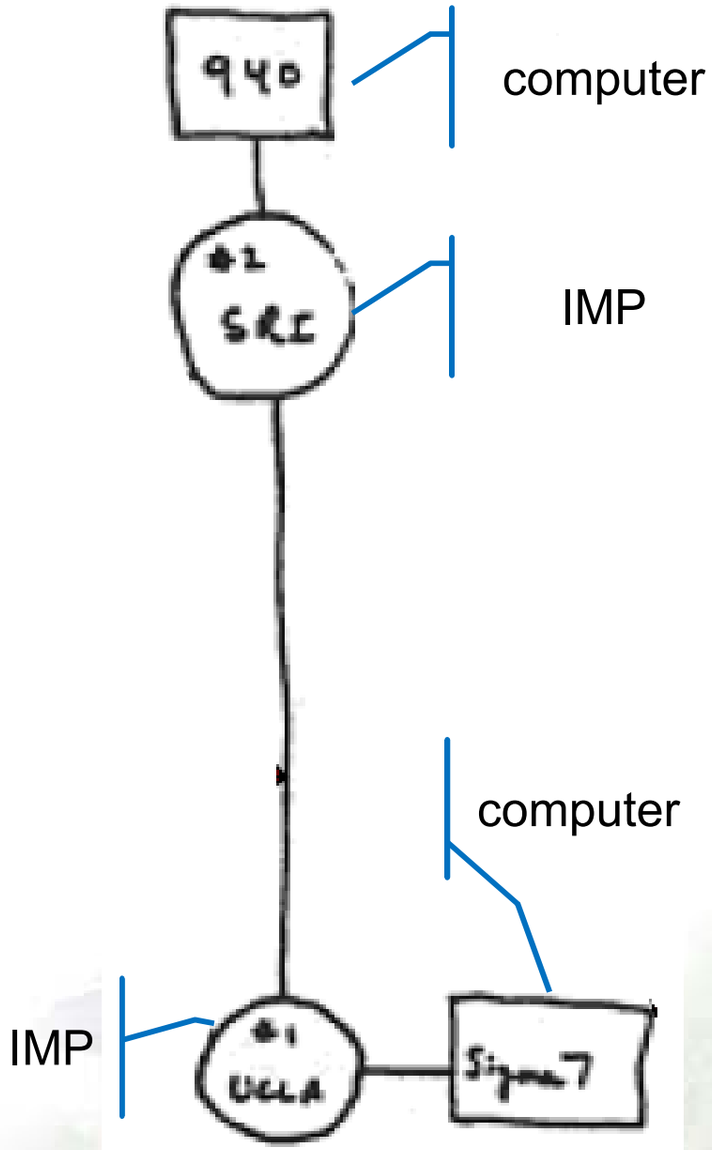
1968, Roberts publishes ARPANET plan



- Roberts decided that the network should start out with four sites:
  - UCLA,
  - the Stanford Research Institute (SRI),
  - the University of Utah,
  - and UC Santa Barbara.
- This would be the core of the later Internet.



- **Connecting Hosts (1940-1972)**
  - **The design of the first network:**
    - **Host computer would be attached to a specialized computer called Interface Message Processor, IMP.**
    - **An IMP would be connected to one another.**
    - **An IMP had to be able to communicate with Attached host and Other IMP**
    - **In August 1969, they delivered the first IMP to UCLA. A month later, the second was delivered to SRI.**
    - **The two were connected and the ARAPNET was born.**



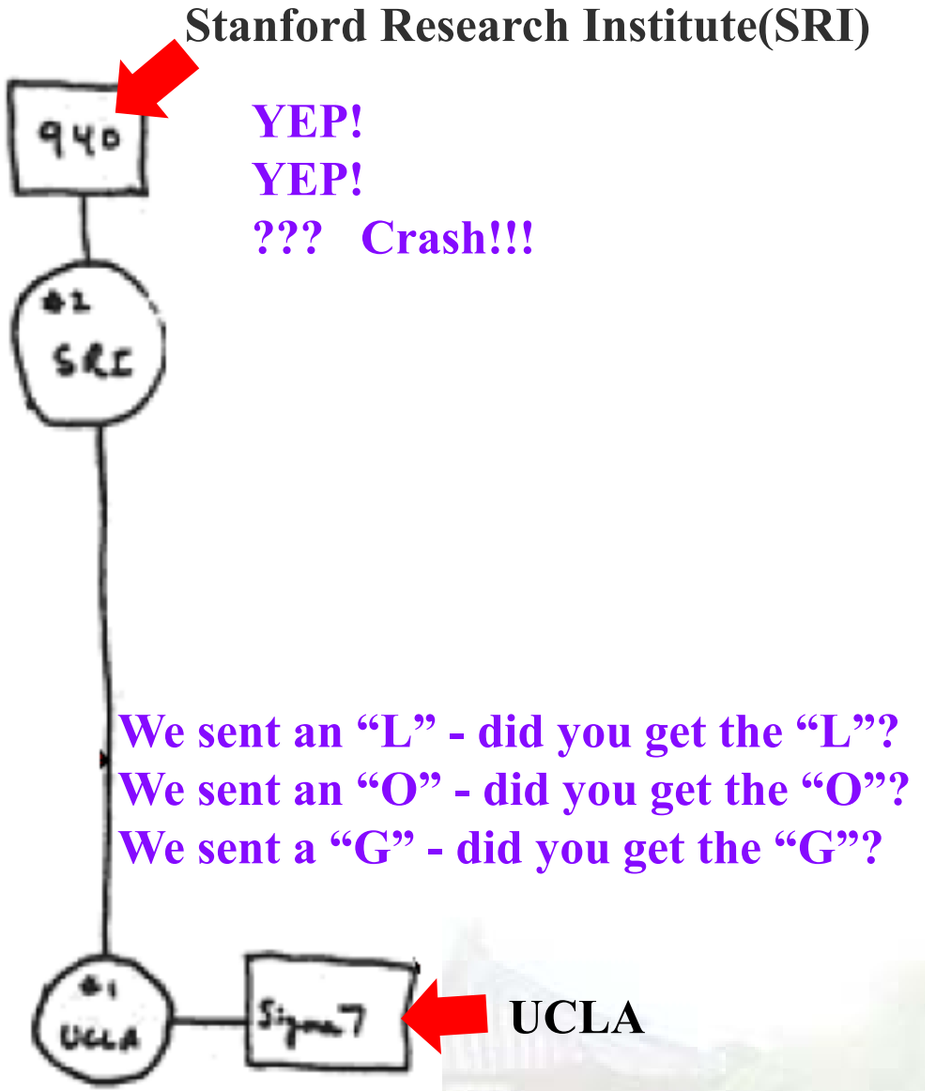
Connecting Hosts (1940-1972)

- On Oct. 29, 1969, Leonard Kleinrock and his graduate student Charley Kline wanted to send a transmission from UCLA's computer to another computer at Stanford Research Institute through ARPANET.
- The message sent by Kleinrock and Kline was intended to be "login." However, the system crashed as soon as they typed the second letter “o”. It took an hour to send the whole word.
- "lo" cemented its place in the Internet's history.
- “‘L’ and ‘O’ is ‘hello,’ and a more succinct, more powerful, more prophetic message we couldn’t have wished for”——Kleinrock



The Internet is BORN at UCLA on October 29, 1969 !

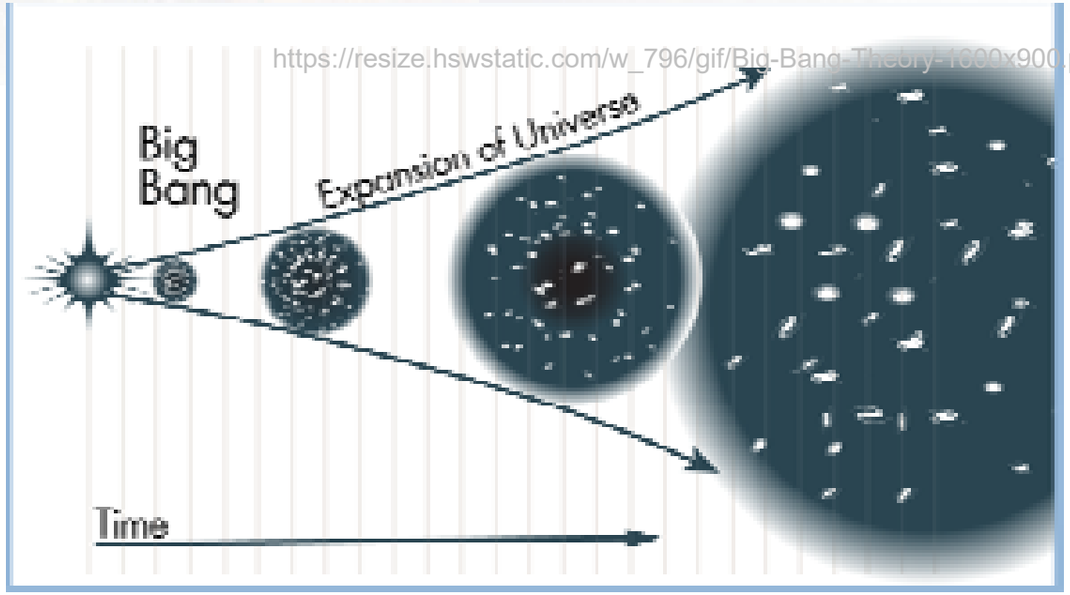
Two years later, in 1971, the first email was sent by MIT researcher Ray Tomlinson — which was also the first time the "@" sign was used to designate a specific recipient of a message.



- **Connecting Hosts (1940-1972)**

***UCLA becomes a harbinger of the modern era as it launches ARPANET, the precursor to the modern internet, from its Measurement Center in September, 1969.**

UCLA becomes the first "node" of the internet, and the first message "LO" of "LOGIN" goes from UCLA to the Stanford Research Institute, Internet Node #2, on October 29th, 1969.



In that moment, the Internet was born, and with it, an almost limitless number of possibilities for the future.



*https://100.ucla.edu/timeline/the-internets-first-message-sent-from-ucla

- **Connecting Hosts (1940-1972)**
  - **1969 Steve Crocker RFC #1 Host-Host Protocol and the NWG**

the 1st RFC: Steve Crocker, "Host Software," known as the Network Control Protocol (NCP), 7th April, 1969.

RFC Index

0001 **Host Software S. Crocker [April 1969]**

50+ years

9654 **Online Certificate Status Protocol (OCSP) Nonce Extension H. Sharma [August 2024]**

https://www.rfc-editor.org/rfc-index.html

Network Working Group — Steve Crocker
Request for Comments: 1 — UCLA
7 April 1969

Title: Host Software
Author: Steve Crocker
Installation: UCLA
Date: 7 April 1969
Network Working Group Request for Comments: 1

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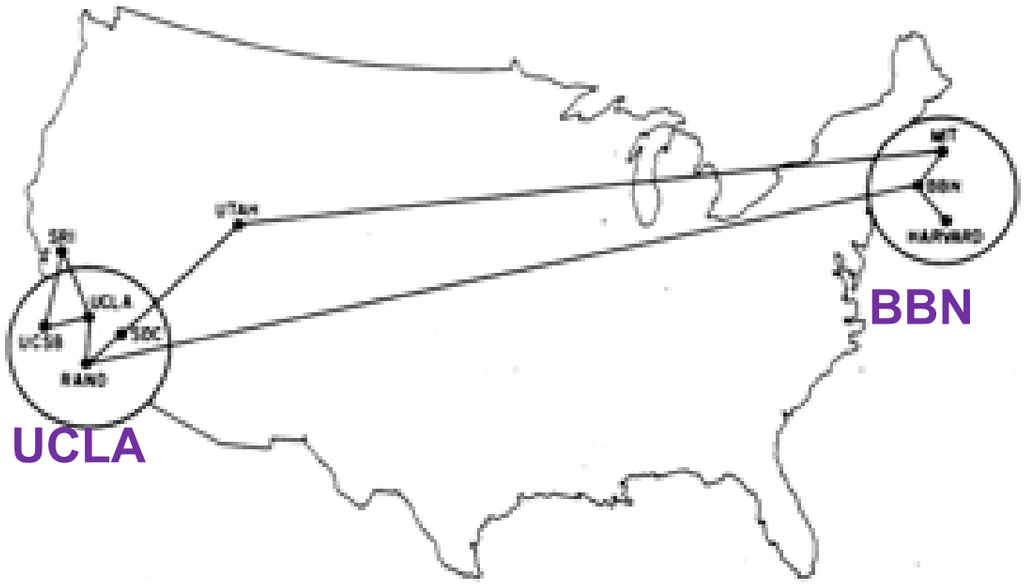
Experiment One

Experiment Two

IETF

Connecting Hosts (1940-1972)

- 1970 ARPANET spans US: UCLA <-> BBN
- 1972 Ray Tomlinson introduce net email and "@"
- 1972 First public demo of ARPANET



- Network Working Group (NWG) adopted a layered approach to the specification of communications protocols.
- A Bhushan, RFC 114, A File Transfer Protocol (April 16, 1971)
- T O'Sullivan, NWG RFC 158, TELNET Protocol (May 19, 1971)

现代网络的发展简史

- Part I Connecting Hosts
- **Part II Connecting Networks**
- Part III Connecting the World
- Part IV Connecting everything

- **Connecting Networks (1972-1989)**
  - **1972** **Additional packet-switching networks besides ARPAnet came into being;**
    - **ALOHAnet, a satellite network linking together universities on the Hawaiian islands;**
    - **Telenet, a BBN commercial packet-switching network based on ARPAnet technology;**
    - **Tymnet; and Transpac, a French packet-switching network.**

The number of networks was beginning to grow.

Once again, with perfect hindsight one might now see that the time was ripe for developing an encompassing architecture for connecting networks together.

- **Connecting Networks (1972-1989)**
  - **1972 Bob Kahn* introduces 4 rules for open-networking architecture.**



    - **Each distinct network had to stand on its own, and no internal changes could be required of any such network before being connected to the Internet.**
    - **Communications would be on a best-effort basis. If a packet didn't make it to the final destination, it could quickly be retransmitted from the source.**
    - **Black boxes (later called gateways and routers) would be used to connect the networks. No information would be retained by the gateways about individual flows of packets passing through them, keeping them simple and avoiding complicated adaptation and recovery from various failure modes.**
    - **There would be no global control at the operations level.**

*** Bob Kahn, along with Vint Cerf, invented the TCP and IP.**

Connecting Networks (1972-1989)

- 1973 Cerf and Kahn design TCP
  - In the spring of 1973, they started by conducting research on reliable data communications across packet radio networks, factored in lessons learned from the Networking Control Protocol, and then created the next generation Transmission Control Protocol (TCP).
  - In the early versions of this technology, there was only one core protocol, which was named TCP (Transmission Control Program).
  - The first version of this predecessor of modern TCP was written in 1973, then revised and formally documented in RFC 675.
  - In 1974, Cerf and Kahn publish TCP specification "A Protocol for Packet Network Intercommunication" [IEEE Trans Comm]





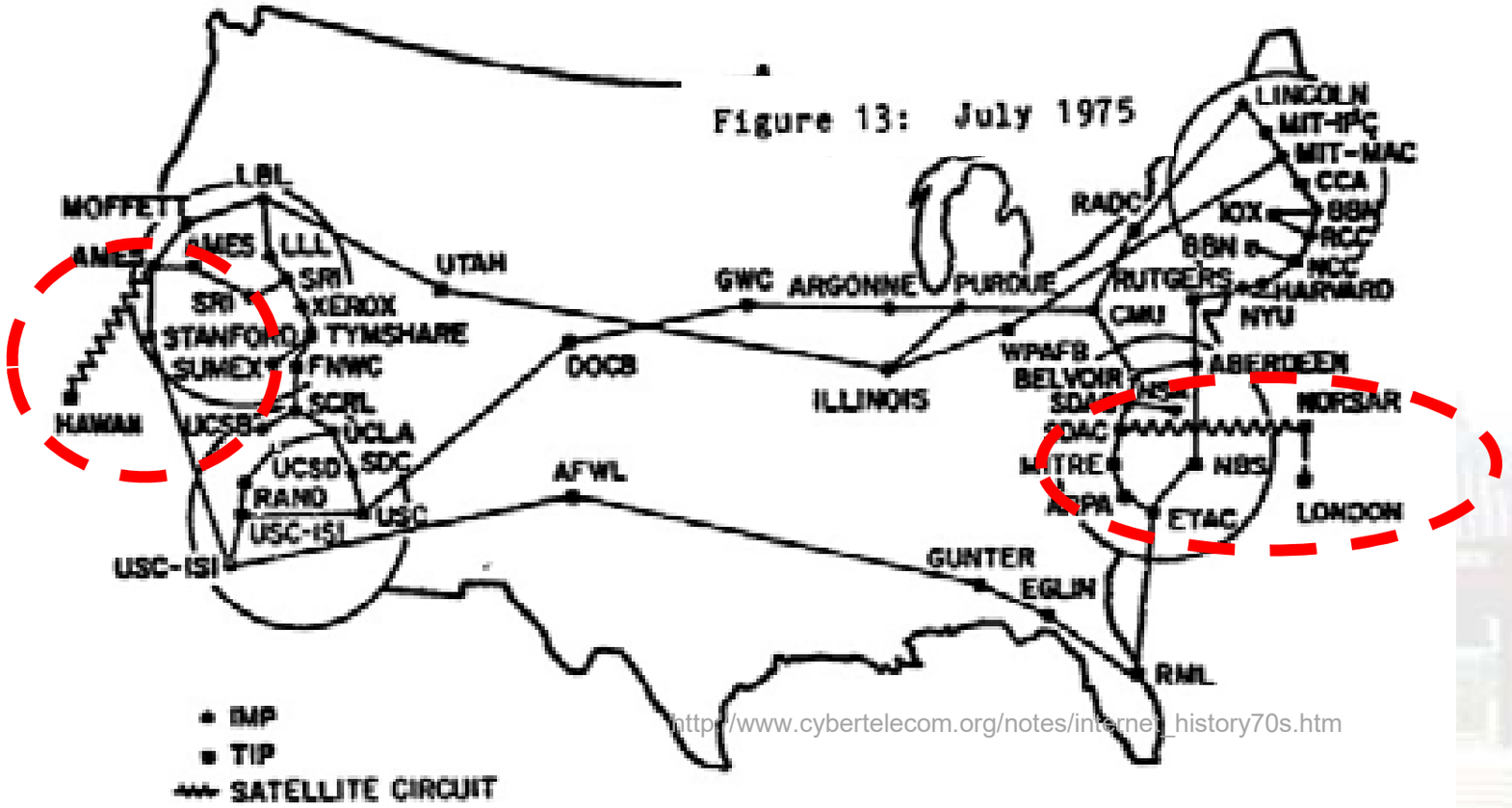
A Protocol for Packet Network Intercommunication

VINTON G. CERF AND ROBERT E. KAHN

INTRODUCTION

Connecting Networks (1972-1989)

- 1973 ARPA deploys SATNET – 1st international connection
  - SATNET: First satellite network on the Internet. Connected Stanford US, University College London England and Norway.

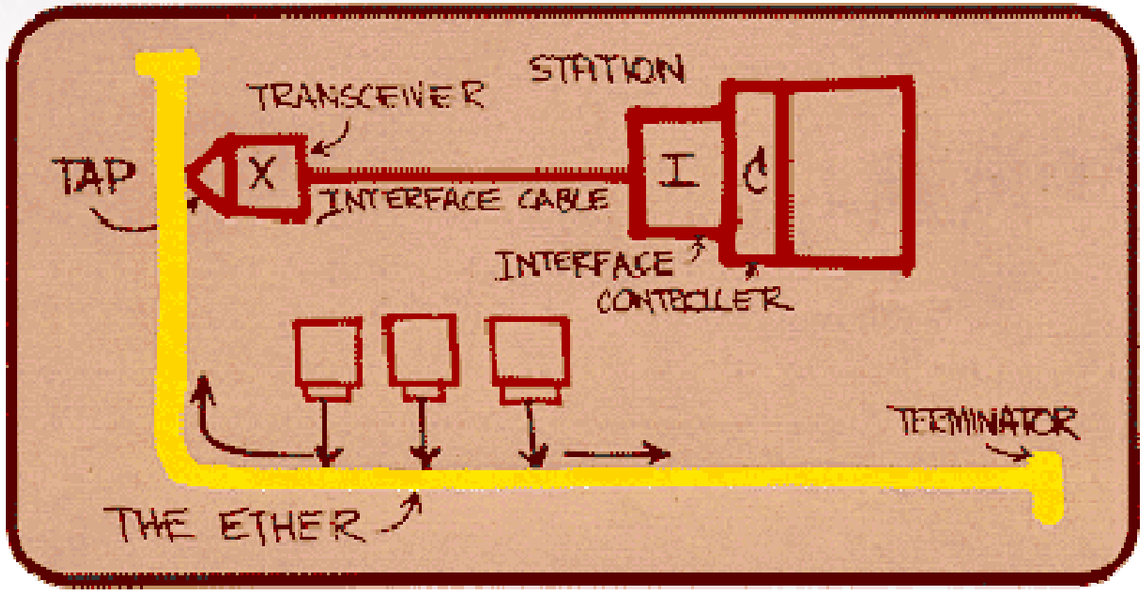


http://www.cybertelecom.org/notes/internet_history70s.htm

Connecting Networks (1972-1989)



- 1973 Bob Metcalfe develops Ethernet idea
  - Metcalfe's Ethernet protocol was motivated by the need to connect multiple PCs, printers, and shared disks together.
  - Each Ethernet local area network was itself a network, and as the number of LANs proliferated, the need to internetwork these LANs together became all the more important.
  - Ethernet technology represented an important step for internetworking.



A 1976 drawing by R. Metcalfe of the Ethernet concept
(from Charles Spurgeon's Ethernet Web Site)



- **Connecting Networks (1972-1989)**
  - 1977 Cerf and Kahn demonstrate interconnection of networks using IP by interconnecting ARPANet, SatNet, Ethernet, and PRNET.
  - 1977 BBN develops first Internet Routers in collaboration with Stanford
  - 1978 TCP splits into TCP and IP driven by Danny Cohen, David Reed and John Schoch to support real-time traffic. This allows the creation of UDP.

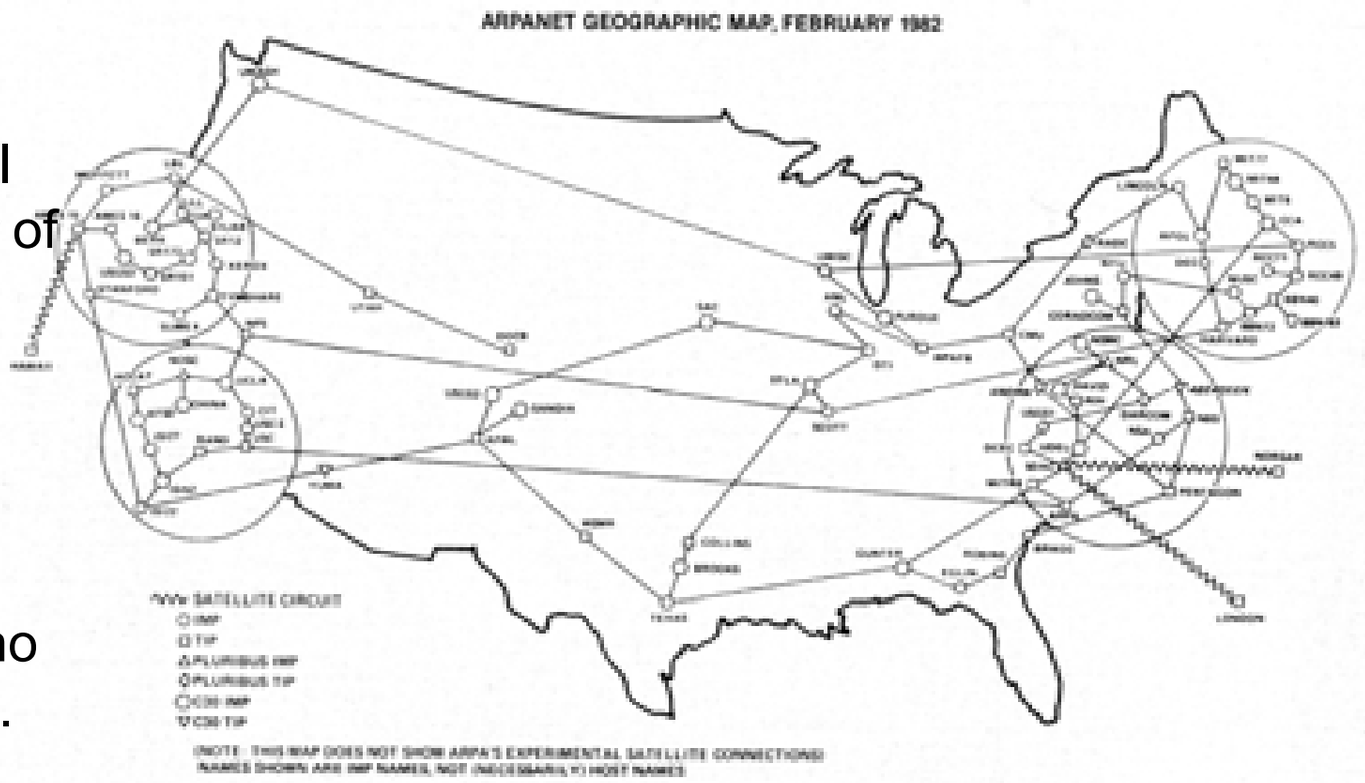
TCPv1; TCPv2; TCP/IPv3 (splitting TCP into TCP and IP), and, in 1978, they settled on IP version4.

IP as originally designed had an eight bit networking field which would be sufficient for at maximum 256 networks



1982: the ARPANET community grows

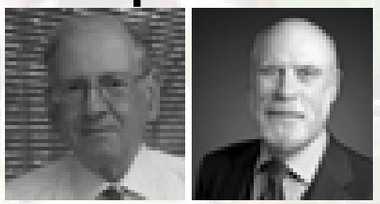
- As the ARPANET entered its second decade, it was still largely confined to the United States.
- Academic institutions depended on federal funding to join the network, so the number of nodes expanded slowly.
- By 1982, the network only had about 100 nodes. But that was enough to support a vibrant online community.
- Long before Facebook and Twitter, ARPANET allowed computer scientists who had access to the network to stay in touch.



- A new bulletin board system called Usenet was invented in 1980 and caught on quickly. Usenet was organized by topic, allowing users to swap programming tips, recipes, jokes, opinions about science fiction, and much more.

1984: ARPANET becomes the internet

- Originally, the entire ARPANET was managed by the military.
- But network operators realized that a centralized network would eventually become unmanageable if it continued to grow.
- They decided that the network should be reorganized as a decentralized "network of networks."
- Under this scheme, different networks would be controlled by different organizations, but all the networks able to communicate using shared standards, forming a shared "internet."
- The military asked the computer scientists Robert Kahn and Vint Cerf to develop new networking standards to make this possible.
- The result was a set of standards known as TCP/IP.



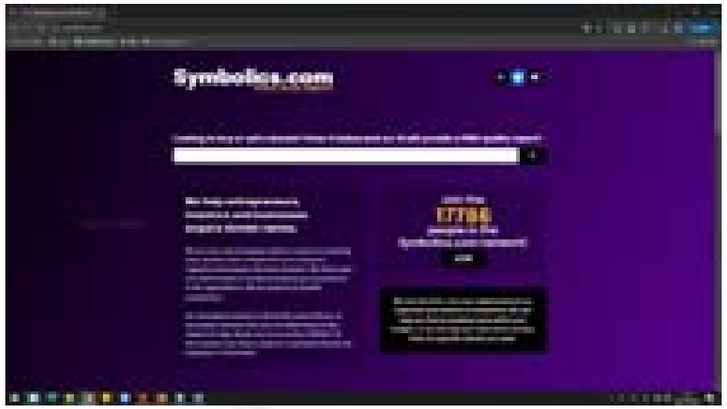
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1984: ARPANET becomes the internet

- These standards specified the basic format of data packets transmitted across the internet.
- On January 1, 1983, the ARPANET switched to using TCP/IP, marking the birth of the modern internet.
- The switch to TCP/IP didn't make much difference from a user perspective — applications like email and Telnet worked about the same as they had before.
- But the new standard paved the way for much faster network growth by lowering the barrier to entry for new networks.
- One of the first new networks to connect to the new internet was CSNET, which was funded by the National Science Foundation to link computer science departments across the country.

Connecting Networks (1972-1989)

- **1985** Paul Mockapetris, Jon Postel and Craig Partridge create the Domain Name system (DNS), the first domain is registered: symbolics.com
- **1985** White House, Science Council (WHSC) Committee on Research in Very High Performance Computing. The C++ was released.
- **1986** Internet Engineering Task Force (IETF) and Internet Research Task Force (IRTF) comes into existence
- **1987** Number of hosts breaks 10,000
- **1987** The first email was sent successfully in China
- **1989** Number of hosts breaks 100,000



It is also worth noting that in the 1980's (and even before), researchers were also developing a "competitor" technology to the Internet architecture. These efforts have contributed to the development of the ATM (Asynchronous Transfer Mode) architecture, a connection-oriented approach based on the use of fixed size packets, known as cells.

Connecting Networks (1972-1989)

- **1990 ARPAnet is terminated on Feb. 28, 1990 and migrated to NSFNET**
- **1991 High-Performance Computing Act that derives the later High Performance Computing and Communications (HPCC) Program**
  - the purpose of this act is to help ensure the continued leadership by the United States and high-performance computing and its applications.
  - The goal of the HPCC Program is to accelerate the development of future generations of high performance computers and networks and the use of these resources in the Federal government and throughout the American economy. Scalable high performance computers, advanced high speed computer communications networks, and advanced software are critical components of a new **National Information Infrastructure** (NII). This infrastructure is essential to our **national competitiveness** and will enable us to strengthen and improve the civil infrastructure, digital libraries, education and lifelong learning, energy management, the environment, health care, manufacturing processes and products, national security, and public access to government information.

现代网络的发展简史

- Part I Connecting Hosts
- Part II Connecting Networks
- **Part III Connecting the World**
- Part IV Connecting everything

NSFNET: The first internet backbone

- During the 1980s, the National Science Network funded several supercomputing centers around the United States. And in 1986 the agency created a TCP/IP-based network called NSFNET to link those supercomputing centers together and allow researchers across the country to use them.
- The primary goal was to allow computer science researchers to log into the supercomputers and perform academic research.
- But NSF decided not to limit NSFNET to that purpose, allowing the network to be used for a wide variety of academic purposes.
- As a result, the NSFNET became the internet's "backbone," the high-speed, long-distance network that allowed different parts of the internet to communicate.
- Schools that didn't have a direct connection to the NSFNET worked together to build regional networks that linked them to each other and to the nearest NSF node.
- This shows the NSFNET as it existed in 1992. By this time, there were 6,000 networks connected to NSFNET, with a third of them located overseas.

Connecting the World (1990-2000)

- **1991** **NSF opens Internet to commercial use**
- **1991** **First Webpage (Tim Berners-Lee)**

On August 6, 1991, without fanfare, British computer scientist Tim Berners-Lee published the first-ever website while working at CERN, the huge particle physics lab in Switzerland.





This computer was used at CERN by British scientist Tim Berners-Lee to devise the World Wide Web (WWW).



Tim Berners-Lee, Inventor of the Web, poses in front of the first World Wide Web Server.

https://www.history.com/news/the-worlds-first-web-site

- **Connecting the World (1990-2000)**
  - **The motivation of the development on the Web**

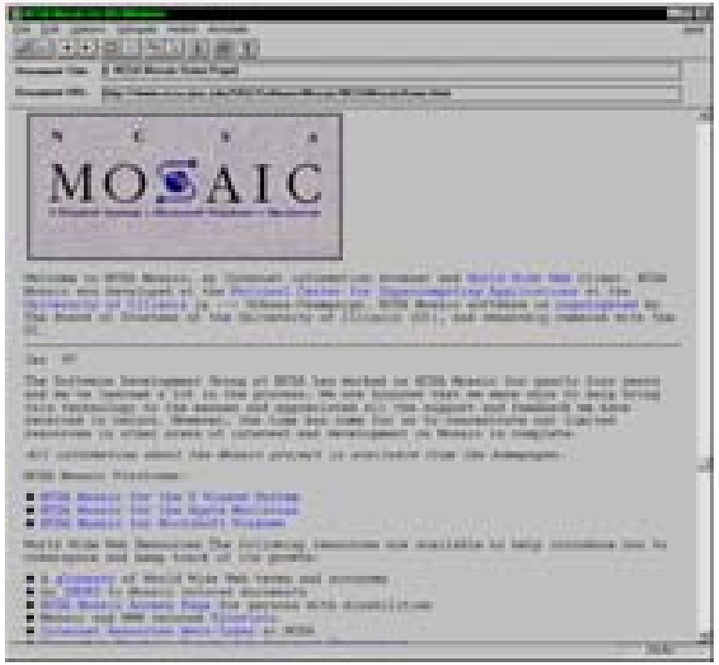


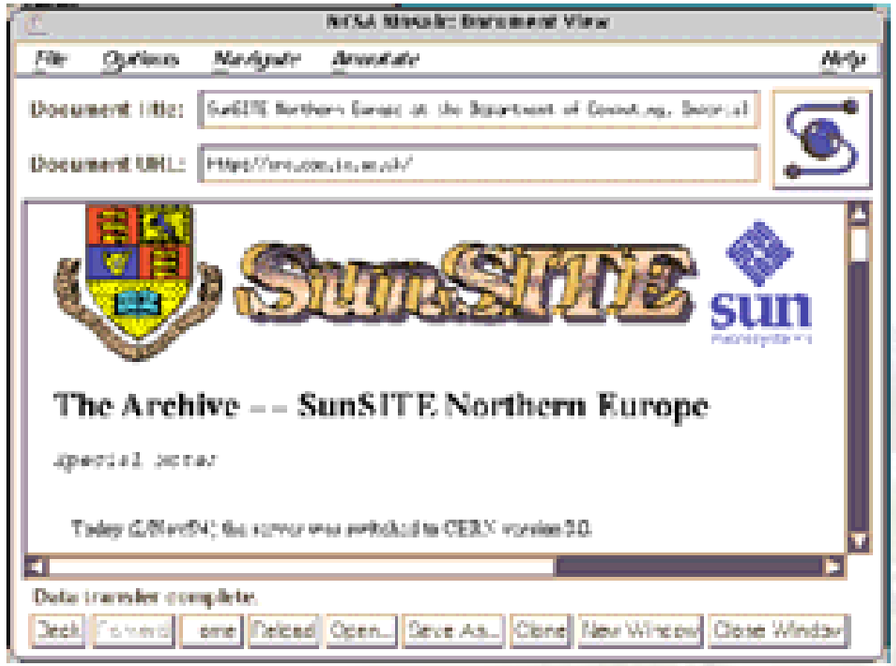
Tim Berners-Lee later stated: "In those days, there was different information on different computers, but you had to log on to different computers to get at it. Also, sometimes you had to learn a different program on each computer."

Berners-Lee didn't try to cash in on his invention and rejected CERN's call to patent his Web technology. He wanted the Web to be open and free so it could expand and evolve as rapidly as possible. As he later said, "Had the technology been proprietary, and in my total control, it would probably not have taken off. You can't propose that something be a universal space and at the same time keep control of it."

http://info.cern.ch/

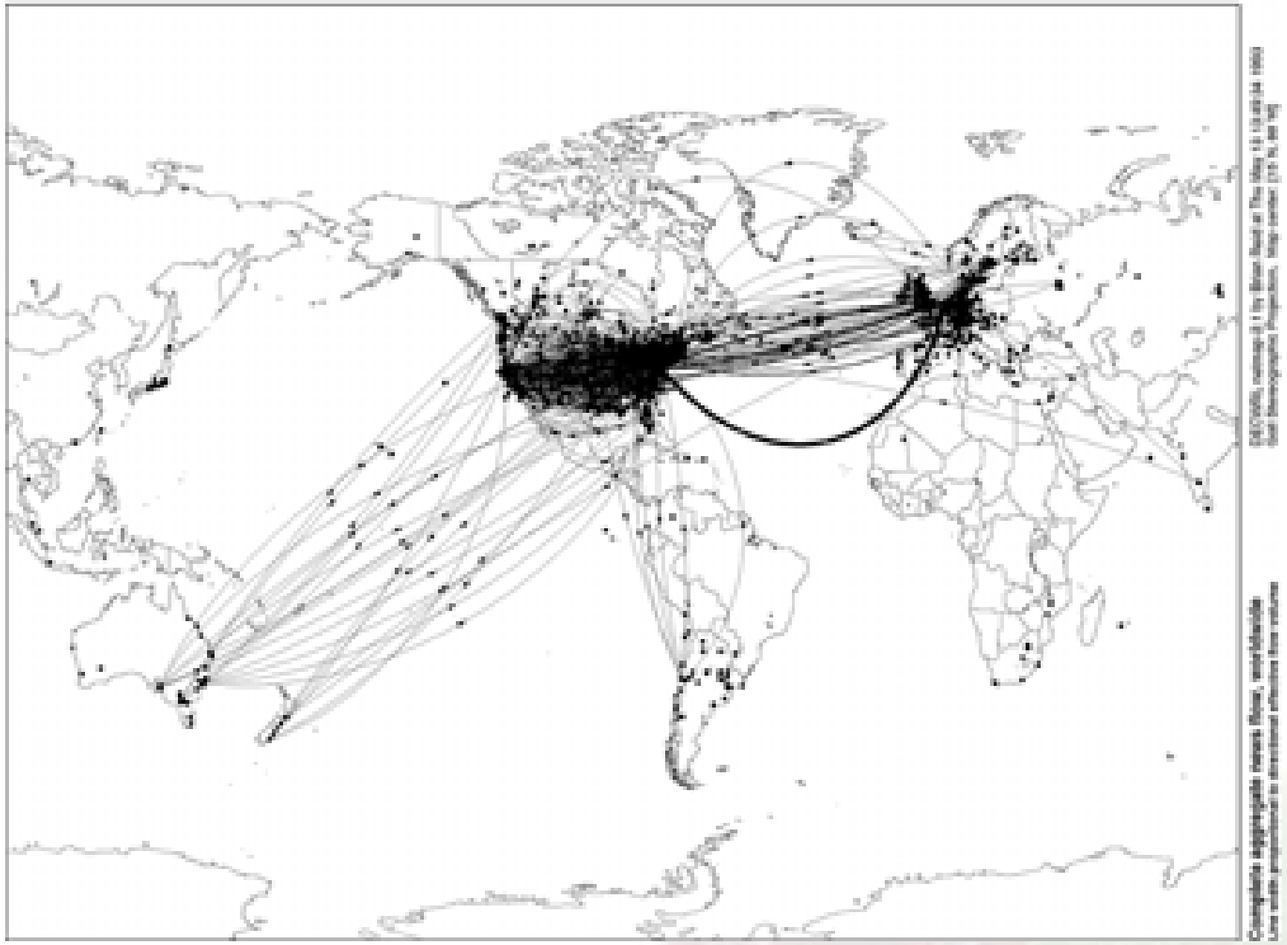
- **Connecting the World (1990-2000)**
  - **1993 Marc Andreeson Mosaic browser**
    - **NCSA Mosaic was neither the first web browser (first was the World Wide Web of Berners-Lee) nor the first graphical web browser (it was preceded by the lesser-known Erwise and ViolaWWW), but it was the web browser credited with popularizing the World Wide Web**
    - **Mosaic was also a client for earlier protocols such as FTP, NNTP, and gopher.**





The internet becomes a global network

- In 1993, the internet was still dominated by the United States but it was becoming a truly global network.
- This is a map of information flow on Usenet, an bulletin board application that allowed users to swap recipes, jokes, programming tips, and more.



1969-1983	Early ARPANET
1983-1990	TCP/IP-based ARPNET
1990-	NSFNET
	Internet

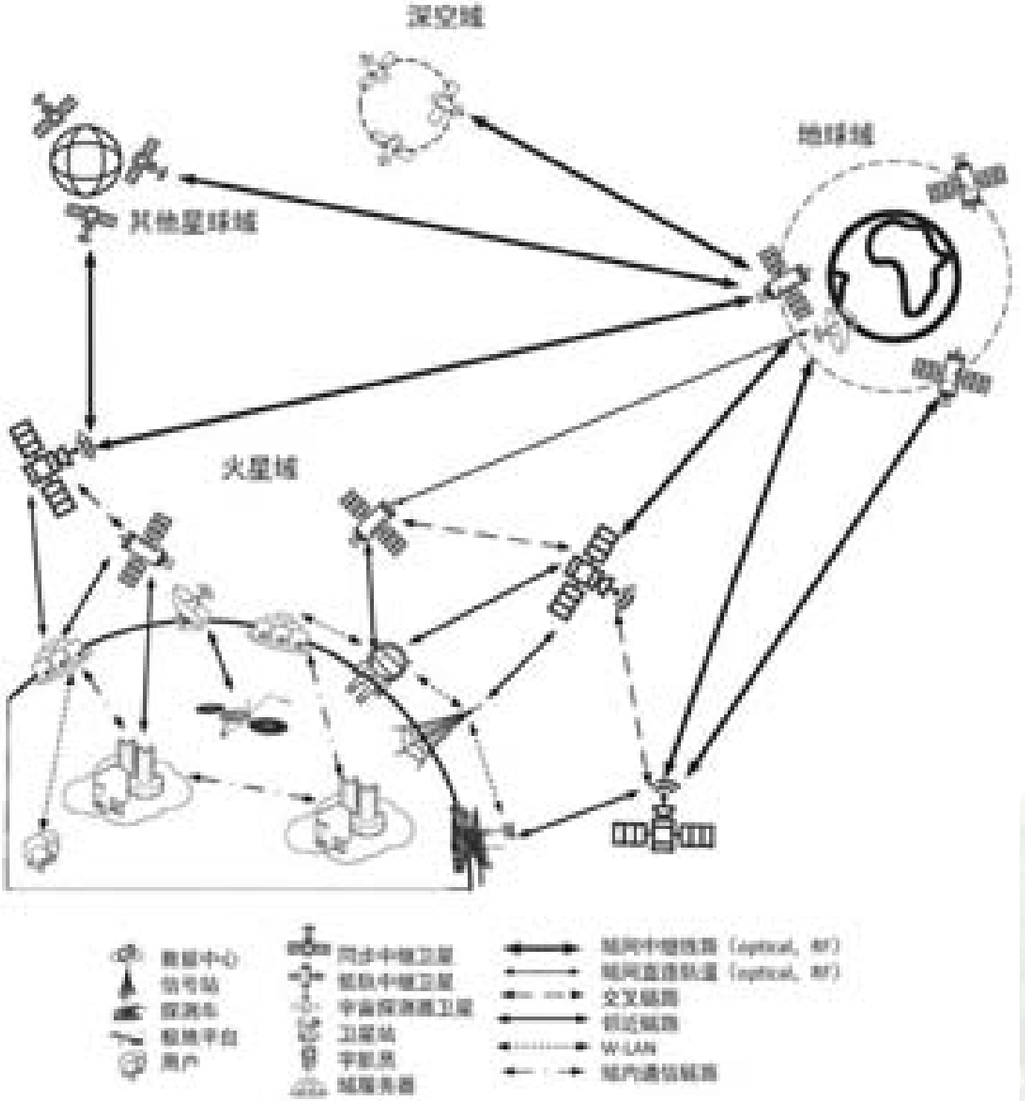
现代网络的发展简史

- Part I Connecting Hosts
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- **Part IV Connecting everything**

Telephone, TCP/IP, Ethernet, Internet, WIFI, 6G, Mobile, Datacenter, Cloud, Fog, IoT …



陆海空天一体化通信网络



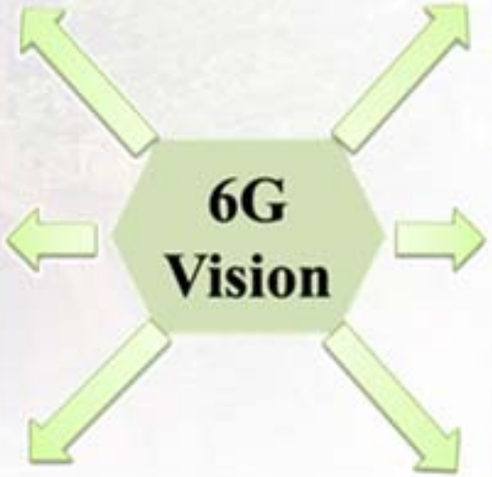
深空通信网络

2030 6G 愿景

Global coverage
- Satellite and UAV communications
- Terrestrial communications
- Maritime communications
- Underwater, underground communications

Full applications
- Integration of communications, computing, storage, control, sensing, localization, robotics, AI, and big data
- Terminal-network-cloud
- Cloud/fog/edge computing

All digital
- Digital twins: mapping between physical world and virtual world
- Intelligent connection of "human-machine-things-environment"



All spectra
- Sub-6 GHz (including short wave and acoustic wave)
- cmWave + mmWave + THz
- Optical wireless

All senses
- Holographic communications/storage
- Truly immersive XR: fusion of virtuality and reality
- Tactile Internet

Strong security
- Physical layer security and network layer security
- Reliable communications
- Intelligent endogenous security

C. -X. Wang *et al.*, "On the Road to 6G: Visions, Requirements, Key Technologies, and Testbeds," in *IEEE Communications Surveys & Tutorials*, vol. 25, no. 2, pp. 905-974, Secondquarter 2023,

6G vision: Global coverage, all spectra, full applications, all senses, all digital, and strong security.

- 6G将与先进计算、大数据、人工智能、区块链等信息技术交叉融合，实现通信与感知、计算、控制的深度耦合，成为服务生活、赋能生产、绿色发展的基本要素。
- 6G将充分利用低中高全频谱资源，实现空天地一体化的全球无缝覆盖，随时随地满足安全可靠的“人机物”无限连接需求。
- 6G将提供完全沉浸式交互场景，支持精确的空间互动，满足人类在多重感官、甚至情感和意识层面的联通交互，通信感知和普惠智能不仅提升传统通信能力，也将助力实现真实环境中物理实体的数字化和智能化，极大提升信息通信服务质量。

- 6G将构建人机物智慧互联、智能体高效互通的新型网络，在大幅提升网络能力的基础上，具备智慧内生、多维感知、数字孪生、安全内生等新功能。 6G将实现物理世界人与人、人与物、物与物的高效智能互联，打造泛在精细、实时可信、有机整合的数字世界，实时精确地反映和预测物理世界的真实状态，助力人类走进人机物智慧互联、虚拟与现实深度融合的全新时代，最终实现“万物智联、数字孪生”的美好愿景。

Networks are changing the world in evolution

A future with endless possibilities is coming …

Are you ready?

Welcome to ComNet!

Thanks

Q & A

Email: xieyi5@mail.sysu.edu.cn
https://cse.sysu.edu.cn/content/2462